

Supplementary Material: One Integer, Three Generations

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This Supplementary Material accompanies the main paper “The Self-Referential Universe: Standard Model Structure from $\tau \otimes \tau = \mathbf{1} \oplus \tau$.” It contains extended proofs, derivation details, and extensions to the dark sector, TQFT, and cosmology. Cross-references to the main text use the form “Section N” or “Eq. (N)”; equations in this supplement are numbered (S1), (S2), etc.

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S1 Notation and Symbol Table

This expands on Section 1 of the main text.

Symbol	Definition	Value
<i>Fundamental constants (from a_1)</i>		
a_1	unique sol. of $a_1! = 4a_1(a_1 + 1)$	5
b_1	$a_1 + 1$	6
ϕ, ϕ'	$(1 \pm \sqrt{a_1})/2$; Galois conjugates	$1.618\dots, -0.618\dots$
N	$a_1! =$ vertices of 600-cell $= 2I $	120
N_{eig}	distinct eigenvalues of the 600-cell adjacency	9
N_{gen}	fermion generations	3
<i>E_8 and group theory</i>		
$h(E_8)$	Coxeter number $= a_1 b_1$	30
$\text{rank}(E_8)$	$N_{\text{eig}} - 1$	8
$2I$	binary icosahedral group	order 120
$2T$	binary tetrahedral group	order 24
A_5	icosahedral rotation group $\cong 2I/\mathbb{Z}_2$	order 60
<i>Coupling constants</i>		
α	fine structure constant; sol. of Eq. (7)	$1/137.036$
α_s	strong coupling $= 1/(2\phi^3)$	0.1180
$\sin^2\theta_W$	Weinberg angle $= b_1/(a_1^2 + 1)$	$6/26$
<i>Mass formula and corrections</i>		
n_f	mass exponent: $m_f = m_e \phi^{n_f}$	integer
(a, b)	$\mathbb{Z}[\phi]$ quantum numbers: $n_f = a_1 a + b_1 b$	
$N(z)$	$\mathbb{Z}[\phi]$ norm: $a^2 + ab - b^2$	integer
C	universal correction coeff. $= 4/(a_1^2 + 1)$	$2/13$
z_{Planck}	Planck exponent $= 4\phi^2$; $\text{Tr} = 12, N = 16$	$10.472\dots$
<i>Internal space</i>		
D_F	finite Dirac operator on $\mathcal{H}_F$	30×30
γ_F	internal chirality $= (-1)^{2j}$ (bipartite)	± 1
γ_{form}	form-degree parity $= (-1)^p$	± 1
σ	Galois automorphism: $\sqrt{5} \rightarrow -\sqrt{5}$	

Table S1: Principal notation. All quantities except m_e (dimensional anchor) are derived from $a_1 = 5$.

S2 Gauge Group: Extended Proofs

This expands on Section 6 of the main text.

S2.1 Algebraic Generation (Burnside)

Proposition S1 (Algebraic generation). *Let $\rho_3 : A_5 \rightarrow SL_3(\mathbb{C})$ be an irreducible 3-dimensional representation. Then:*

- (i) $\text{Span}_{\mathbb{C}}\{\rho_3(g) : g \in A_5\} = M_3(\mathbb{C})$ (Burnside's theorem).

- (ii) *In particular, $\mathfrak{sl}_3(\mathbb{C}) = \text{Lie}(SL_3(\mathbb{C}))$ is contained in this span: every infinitesimal $SU(3)$ gauge transformation is a $\mathbb{C}$ -linear combination of the 60 discrete icosahedral symmetry operations.*

Proof. By Burnside’s theorem (a corollary of the Artin–Wedderburn structure theorem), if $\rho : G \rightarrow GL(V)$ is irreducible for a finite group G , then $\rho(\mathbb{C}[G]) = \text{End}(V)$. Applied to ρ_3 with $\dim V = 3$: $\text{Span}_{\mathbb{C}} \rho_3(A_5) = M_3(\mathbb{C})$, which has dimension 9. The traceless part has dimension $8 = \dim(\mathfrak{sl}_3)$, confirming (ii). Both claims are verified computationally: the 60×9 matrix of vectorized rotation matrices has rank 9. $\square$

Remark. The Zariski closure of a finite subgroup $\Gamma \subset GL_n(\mathbb{C})$ is Γ itself (finite sets are Zariski closed). The bridge from discrete to continuous symmetry therefore does *not* proceed via algebraic-geometric closure. Instead, three complementary mechanisms operate: (1) the representation-theoretic decomposition $12 = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ *identifies* the gauge group uniquely (Theorem 6 in the main text); (2) Burnside’s theorem shows the discrete symmetries *algebraically span* the full gauge Lie algebra (Proposition S1); (3) the spectral action (Section 9 of the main text) provides the Yang–Mills *dynamics* with continuous gauge invariance.

Remark (Distler–Garibaldi). The impossibility theorem of Distler and Garibaldi [1] proves that three chiral SM generations cannot be embedded as representations of E_8 . This theorem does *not* apply to the present framework: the gauge group $SU(3) \times SU(2) \times U(1)$ is identified via the A_5 decomposition of the vertex figure, and fermions arise from the nine irreducible representations of $2I$ —not from E_8 representations. The E_8 structure enters exclusively through the McKay correspondence (graph topology and spectral action coefficients), not through direct embedding of fermion representations.

S2.2 Extended Chirality Discussion

The full proof that the McKay bipartite grading $\gamma_F = (-1)^{2j}$ yields the Standard Model chiral gauge coupling is presented here.

The McKay graph of $2I$ is a tree (9 nodes, 8 edges), hence bipartite. The unique bipartition assigns chirality $\gamma_F = +1$ (WHITE) to the five integer-spin irreps

$$\{\rho_1(1), \rho_4(3), \rho_5(3), \rho_6(4), \rho_8(5)\}$$

with total dimension 16, and $\gamma_F = -1$ (BLACK) to the four half-integer-spin irreps

$$\{\rho_2(2), \rho_3(2), \rho_7(4), \rho_9(6)\}$$

with total dimension 14.

The operator γ_F satisfies $\gamma_F^2 = 1$ and $\{\gamma_F, D_F\} = 0$ exactly (a consequence of the tree/bipartite structure: every McKay edge connects WHITE to BLACK). The representation ring $\text{Rep}(2I)$ respects the $\mathbb{Z}/2$ grading: $\text{WHITE} \otimes \text{WHITE} = \text{WHITE}$, $\text{BLACK} \otimes \text{BLACK} = \text{WHITE}$, $\text{WHITE} \otimes \text{BLACK} = \text{BLACK}$ (verified for all $9 \times 9 = 81$ tensor products using A_5 decomposition).

Taking $T_3(\text{WHITE}) = -\frac{1}{2}$ and $T_3(\text{BLACK}) = +\frac{1}{2}$ reproduces the SM left-handed weak isospin $T_{3,L}$ for 8 out of 9 fermions; the single failure (s quark) is structural, arising from the μ - s degeneracy at the $w = 0$ McKay edge. The $SU(2)$ Casimir sum $\sum C_2 = 26 = a_1^2 + 1$ connects γ_F to the Weinberg angle.

S3 Coupling Constants: QCD Running and Energy Scale

This expands on Section 7 of the main text.

S3.1 QCD Running Consistency Check

Standard 3-loop QCD running from the framework value $\alpha_s(M_Z) = 1/(2\phi^3) = 0.1180$ reproduces the measured strong coupling at all accessible scales:

Scale	Framework (3-loop)	PDG	Error
$m_t = 172.7$ GeV	0.1077	0.1085	0.8%
$m_b = 4.18$ GeV	0.2245	0.2268	1.0%
3 GeV	0.2528	0.2530	0.1%
$m_\tau = 1.78$ GeV	0.3177	0.332	4.3%
$m_c = 1.27$ GeV	0.3832	0.392	2.3%

The errors are consistent with those obtained by running from the PDG world-average $\alpha_s(M_Z) = 0.1179$ using the same 3-loop beta function with flavor thresholds at m_c , m_b , m_t . The 4% deviation at m_τ reflects the well-known limitation of perturbative QCD at low scales, not a deficiency of the framework. At $\mu \approx 2 \times 10^{16}$ GeV, the framework gives $1/\alpha_3 \approx 46$ and $1/\alpha_2 \approx 46$ (1-loop electroweak running), suggesting near-unification of $SU(3)$ and $SU(2)$ with a gap of only 0.7%.

S3.2 Energy Scale Discussion

The framework predicts fixed algebraic values, not running couplings. We compare $\alpha_s = 1/(2\phi^3)$ with $\alpha_s(M_Z)$ and $\sin^2 \theta_W = 6/26$ with $\sin^2 \theta_W(M_Z)$ because the 600-cell’s characteristic scale is the electroweak scale: $m_Z = m_e \phi^{a_1^2} \cdot \alpha(m_Z)/\alpha(0)$ (Section 10 of the main text), so the “lattice spacing” of the 600-cell corresponds to M_Z^{-1} . This identification is self-consistent—the spectral conditions selecting α_s use eigenvalue multiplicities of the 600-cell, which is the same object whose mass spectrum yields m_Z —but the reason *why* the framework naturally produces M_Z -scale values rather than, say, Planck-scale values is not derived from first principles.

Three number fields. Each coupling lives in a different algebraic structure: $\sin^2 \theta_W = 6/26 \in \mathbb{Q}$ (rational, from graph combinatorics), $\alpha_s = 1/(2\phi^3) \in \mathbb{Z}[\phi]$ (algebraic, from spectral conditions), and α satisfies a quadratic involving π (transcendental, from the Hopf holonomy). The hierarchy $\mathbb{Q} \subset \mathbb{Z}[\phi] \subset \mathbb{Z}[\phi, \pi]$ reflects the decreasing discreteness of each gauge sector. With GUT normalization $\alpha_1^{\text{GUT}} = \frac{5}{3}\alpha/(1 - \sin^2 \theta_W)$, the relation $1/\alpha_1^{\text{GUT}} = 2/\alpha_2$ holds *exactly* for $a_1 = 5$, since $3(a_1^2 - a_1)/(5b_1) = 60/30 = 2$.

S4 Generation Theorem: Independent Confirmations

This expands on Section 7 of the main text.

Five additional routes yield $N_{\text{gen}} = 3$:

(i) *Soliton spectral splitting*—a rank-3 perturbation splits each eigenvalue group into exactly 3 sub-levels with universal pattern $(m_k - 3, 2, 1)$.

(ii) *Representation branching*—the 6-dimensional irrep ρ_8 of $2I$ restricts to $2T$ as $\psi_3 \oplus \psi_4 \oplus \psi_5$, the three 2-dimensional irreps.

(iii) *Vertex decomposition*: $96 = 16 \times 3 \times 2$.

(iv) *Nyquist sampling on the Hopf base*. The Hopf fibration $S^3 \rightarrow S^2$ projects the 600-cell onto an icosahedron with 12 vertices on S^2 . The spherical harmonics Y_l^m on S^2 , truncated by the Nyquist criterion $(l_{\max} + 1)^2 \leq 12$, give $l_{\max} = 2$ (since $9 \leq 12$ but $16 > 12$), hence $N_{\text{gen}} = l_{\max} + 1 = 3$. The generation Casimir $g(g+1) = l(l+1)$ is the S^2 Laplacian eigenvalue. Remarkably, the highest mode $l = 2$ has dimension $2l + 1 = 5 = a_1$ and Casimir $l(l+1) = 6 = b_1$: the framework’s two fundamental constants are the dimension and Casimir of the *same* spin-2 representation of $SU(2)$.

(v) *Chirality and $\mathbb{Z}[\phi]$ units*. Among lines with odd form-degree parity ($\gamma = -1$, fermionic), $a = 1$ is the unique line admitting multiple $\mathbb{Z}[\phi]$ units. The Fibonacci sequence has exactly three terms equal to 1 ($F(m) = 1$ for $m \in \{-1, 1, 2\}$), giving $b \in \{0, 1, 2\}$.

Negative result: anomaly cancellation. Standard anomaly cancellation ($\text{Tr}(Y) = 0$, $\text{Tr}(Y^3) = 0$, Witten’s $SU(2)$ global anomaly) is satisfied per generation and places no constraint on N_{gen} . The $\mathbb{Z}[\phi]$ unit argument remains the primary derivation.

S5 Fermion Masses: Complete Derivation Chain

This expands on Section 8 of the main text.

S5.1 L1-Minimality: Unique (a, b) from n

Given a mass exponent n , the quantum numbers (a, b) are the unique solution of:

$$(a, b) = \arg \min_{5a+6b=n} (|a| + |b|) \quad (\text{S1})$$

This is a topological complexity minimization: fermions “choose” the simplest path on the 600-cell graph. It has been verified computationally for all 9 fermions, with a unique minimizer in each case.

S5.2 Algebraic Structure: Units and Primes

The algebraic element $z_f = a + b\phi \in \mathbb{Z}[\phi]$ encodes the fermion’s position in the number ring. Its norm $N(z) = a^2 + ab - b^2$ reveals a clean partition:

Fermion	z_f	$ N(z) $	Status in $\mathbb{Z}[\phi]$
μ, τ, u, d, s	various	1	Units ($ N = 1$)
e	$\phi^0 = 1$	1	Ground state (trivial unit)
Charm	$\phi\sqrt{a_1}$	$a_1 = 5$	Ramification prime
Top	$(a_1 - 1) + \phi$	19	Split prime
Bottom	$-1 + (a_1 - 1)\phi$	19	Galois partner of top

Six of nine fermions are units of $\mathbb{Z}[\phi]$. The three non-units (charm, top, bottom) involve the two primes that ramify or split in $\mathbb{Z}[\phi]$: $5 = a_1$ (ramified) and $19 = a_1^2 - a_1 - 1$ (split).

S5.3 Exponent Sum Rules and Galois Exponents

The mass exponents satisfy remarkable identities linking quark pairs to E_8 invariants:

$$n_u + n_d = 3 + 5 = 8 = \text{rank}(E_8) \quad (\text{S2})$$

$$n_\tau + n_u = 17 + 3 = 20 = 4a_1 \quad (\text{coefficient in the } \alpha \text{ equation}) \quad (\text{S3})$$

$$n_\tau + n_b = 17 + 19 = 36 = b_1^2 \quad (\text{S4})$$

$$n_t + n_b = 26 + 19 = 45 = a_1 \cdot N_{\text{eig}} \quad (\text{S5})$$

Each fermion $z = a + b\phi$ has a Galois conjugate exponent $n' = a_1 a - b$. A remarkable fermion-specific identity: for the charm quark, $n \cdot n' = 16 \times 9 = 144 = 12^2 = (\text{vertex degree})^2$.

S5.4 Uniqueness Theorem: Full Proof

Theorem S1 ($\mathbb{Z}[\phi]$ Norm Selection). *Let $n_f \in \{0, 3, 5, 11, 11, 16, 17, 19, 26\}$ be the mass exponents. For each fermion f , write $n_f = 5a_f + 6b_f$ and define $z_f = a_f + b_f\phi \in \mathbb{Z}[\phi]$. The $\mathbb{Z}[\phi]$ -valued function $f \mapsto z_f$ on the McKay graph $\hat{E}_8$ is **uniquely** determined by:*

(C1) **Irreducibility.** *Each z_f is zero, a unit, or an irreducible element of $\mathbb{Z}[\phi]$. Equivalently, $|N(z_f)| \in \{0, 1\} \cup \{p \text{ prime} : p \equiv 0, \pm 1 \pmod{a_1}\}$.*

(C2) $\mathbb{Z}[\phi]$ **flatness.** *The total edge norm on the McKay tree vanishes against the node sum:*

$$\sum_{\text{edges}(i,j)} N(z_i - z_j) = -b_1. \quad (\text{S6})$$

Then the assignment is **unique**:

$$(a, b) = (0, 0), (3, -2), (1, 0), (1, 1), (1, 1), (2, 1), (1, 2), (-1, 4), (4, 1).$$

Proof. The electron has $n_e = 0$, giving $N(z_e) = -19t_e^2$; constraint (C1) forces $t_e = 0$. The main text (Theorem ??) gives a constructive derivation: the main-chain weights are forced by minimal local edge lifts in $\mathbb{Z}[\phi]$, the branch labels are resolved by affine- E_8 geometry and chirality, and the prime-sector assignment is closed by the Galois-pair relation $z_b = \phi \sigma(z_t)$. No bounded search is needed.

As an independent *computational cross-check*, the binary quadratic form with discriminant $a_1 = 5$ admits at most 20 integer solutions t per fermion in $|t| \leq 15$ (the leading term $-19t^2/25$ ensures $|N(z)| > 19$ for $|t| > 5$, so $|t| \leq 15$ is ultra-conservative). The total number of combinations satisfying (C1) is $13 \times 17 \times 20 \times 20 \times 12 \times 16 \times 4 \times 8 = 543,129,600$. A tree-decomposition search verifies that *exactly one* of these $\sim 5.4 \times 10^8$ combinations satisfies the edge sum constraint (C2), confirming the constructive result. $\square$

The norm set is a consequence. The unique solution has $|N(z_f)| \in \{0, 1, a_1, a_1^2 - a_1 - 1\} = \{0, 1, 5, 19\}$. The second non-trivial value, $19 = a_1^2 - a_1 - 1$, is not an input: it *emerges* as the unique split prime compatible with the edge flatness condition.

Derived identities. The following are all consequences of the unique solution:

$$\begin{aligned} \sum_f N(z_f) &= +b_1 = 6, & \sum_{\text{edges}} N(z_i - z_j) &= -b_1 = -6, \\ \sum_f |N(z_f)| &= \text{Tr}(A^4) = 48, & \prod_{N \neq 0} N(z_f) &= -a_1 \cdot 19^2 = -1805. \end{aligned} \quad (\text{S7})$$

S5.5 Norm-Log Corrections: Full Derivation

Why the logarithm of the norm: Arakelov height. The appearance of $\ln |N(z)|$ is a consequence of the arithmetic of $\mathbb{Z}[\phi]$. For an algebraic integer $z = a + b\phi$ in the ring of integers of $\mathbb{Q}(\sqrt{5})$, the *Arakelov height* is

$$h(z) = \frac{1}{[\mathbb{Q}(\sqrt{5}) : \mathbb{Q}]} \sum_{v|\infty} \ln |z|_v = \frac{1}{2} (\ln |z| + \ln |\sigma(z)|) = \frac{1}{2} \ln |N(z)| \quad (\text{S8})$$

The mass correction receives contributions from *both* places—the physical sector ($|z|$) and the Galois conjugate sector ($|\sigma(z)|$)—giving $\delta_q \propto \ln |z| + \ln |\sigma(z)| = \ln |N(z)|$.

The universal coefficient. The proportionality constant is $C = 4/(a_1^2 + 1) = d_{\text{ST}}/(a_1^2 + 1) = 2/13$, the ratio of the spacetime dimension $d_{\text{ST}} = a_1 - 1 = 4$ to the modular invariant $a_1^2 + 1 = 26$. Equivalently, $C = \text{rank}(E_8)/(2(a_1^2 + 1))$, since $\text{rank}(E_8) = 2 d_{\text{ST}} = 8$. Both $\sin^2 \theta_W = b_1/(a_1^2 + 1)$ and $N_{\text{gen}} = 3$ are derived from a_1 , so $C = 2 \sin^2 \theta_W / N_{\text{gen}}$ is algebraically determined with zero free parameters.

The three down-quark sectors. The three down-quark cases correspond to the three algebraic sectors of $\mathbb{Z}[\phi]$, distinguished by the Artin product formula: (i) the *rational* sector ($b = 0, z \in \mathbb{Z}$): the Arakelov height vanishes ($\ln |N(1)| = 0$), and σ acts trivially on $z_d = 1$. However, σ acts non-trivially on the *representation*: ρ_2 (irrep **3** of A_5) has Galois conjugate ρ_8 (irrep **3'**). The mass correction equals the character cross-term on the Galois-active sector (5-cycles of A_5 , the only elements where $\chi_{\mathbf{3}} \neq \chi_{\mathbf{3}'}$):

$$\delta_d = \langle \chi_{\mathbf{3}}, \chi_{\mathbf{3}'} \rangle_{5\text{-cyc}} = \frac{24 N(\phi)}{|A_5|} = \frac{2}{a_1} \cdot (-1) = -\frac{2}{5} \quad (\text{S9})$$

The fraction $24/60 = 2/a_1$ counts 5-cycles via Sylow theory ($n_{\text{Syl}} = b_1 = 6$ subgroups, each with $a_1 - 1 = 4$ non-identity elements), and $N(\phi) = \phi \cdot \phi' = -1$. When z is rational, the Galois correction shifts from exponent space (Arakelov, Morrey) to representation space (character inner product); (ii) the *unit* sector ($|N| = 1, b \neq 0$): the Arakelov height vanishes but the Galois anti-symmetric Cayley eigenvalue $|G(\rho_4)| = 3 = N_{\text{gen}}$ (Proposition S8) is non-zero, giving $\delta_s = -N_{\text{gen}} C / \phi^2$; (iii) the *prime* sector ($|N| > 1$): both valuations are non-trivial, and $-C \ln |N| / \phi$ is the Arakelov height with Galois suppression.

Lepton corrections: the Morrey–Sobolev mechanism. The exponent $k = 3/4$ is derived from the Morrey embedding. By Morrey’s inequality, the Sobolev space $W^{1,d}(\mathbb{R}^1)$ embeds continuously into the Hölder space $C^{0,1-1/d}(\mathbb{R}^1)$. With $d = d_{\text{ST}} = \text{Tr}(\phi^3) = 4$, the Hölder exponent is $1 - 1/4 = 3/4$.

The saturation argument proceeds in three steps: (1) summability fixes d ; (2) the spectral triple is d_{ST} -summable; (3) for leptons (WHITE nodes), $G_k = 0$ removes all spectral constraints beyond regularity, forcing the correction to the extremal Morrey function $f^*(z') = C|z'|^{1-1/d}$.

Derivation of the coefficient. The lepton coefficient is uniquely determined:

$$c_\ell = \frac{C \cdot \phi^3}{\text{Tr}(\phi^3)} = \frac{C \phi^3}{4} \quad (\text{S10})$$

where $\text{Tr}(\phi^3) = 4 = \dim(\text{spacetime})$.

S5.6 The Galois Cascade: Completeness and Cohomological Structure

This section provides the completeness proof and cohomological interpretation of the Galois cascade (Section 8.5 of the main text).

Completeness. The nine fermions partition into the cascade levels as follows:

Level	Condition	Fermions	Origin
0	$z = 0$	e	input
1	$ N(z) > 1$	c, t, b	symmetric Arakelov
2a	$ N \leq 1, b \neq 0, G_k = 0$	μ, τ	anti-sym. + Morrey
2b	$ N \leq 1, b \neq 0, G_k > 0$	s	anti-sym. + spectral
2c	$ N \leq 1, b \neq 0, G_k < 0$	u	anti-sym., suppressed
3	$b = 0, z \neq 0$	d	representation pairing

Each fermion appears in exactly one row. The partition is exhaustive and mutually exclusive: the conditions $\{z = 0\}$, $\{|N| > 1\}$, $\{|N| \leq 1, b \neq 0\}$, $\{b = 0, z \neq 0\}$ cover all of $\mathbb{Z}[\phi]$.

Cohomological interpretation. The Galois group $\text{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q}) = \mathbb{Z}/2 = \{1, \sigma\}$ acts on the Wilson line $W_f = (z_f, \rho_f)$. For a $\mathbb{Z}/2$ -module M , the Tate cohomology group $\hat{H}^1(\mathbb{Z}/2, M) = \ker(\text{Norm})/\text{im}(\sigma - 1)$ measures the obstruction to Galois invariance. The three cascade levels correspond to a filtration of $\hat{H}^1(\mathbb{Z}/2, W)$:

- *Level 1:* the symmetric part $h(z) = \frac{1}{2} \ln |N(z)|$ (Arakelov height). Non-zero only when $|N(z)| > 1$.
- *Level 2:* the anti-symmetric part $|z'| = |\sigma(z)|$, active when $h(z) = 0$ but $\sigma(z) \neq z$. The sub-cases 2a/2b/2c are distinguished by the McKay spectral data G_k (the Galois anti-symmetric Cayley eigenvalue), which determines whether the obstruction propagates to the mass correction.
- *Level 3:* the representation component $\langle \chi_\rho, \chi_{\sigma(\rho)} \rangle_{5\text{-cyc}}$, active when $\sigma(z) = z$ (rational exponent). The Galois obstruction shifts entirely to the representation space.

Why the cascade is ordered. For fermions where Level 1 is active ($|N| > 1$), the representation cross-term $\langle \chi, \chi_\sigma \rangle_{5\text{-cyc}} = \pm 2/a_1$ is also non-zero (the t and b quarks sit at nodes ρ_7, ρ_8 with $\sigma(\rho) \neq \rho$). However, the Arakelov contribution $C \ln |N| = C \ln 19 \approx 0.45$ dominates over the representation term $2/a_1 = 0.40$ by a factor of $\ln 19/(2/(C a_1)) = C \ln 19/(2/a_1) \approx 1.13$. More precisely, the representation cross-term for t and b is absorbed into the Arakelov mechanism: the full character orthogonality of A_5 ensures $\langle \chi, \chi_\sigma \rangle_{\text{full}} = 0$, so the 5-cycle cross-term is exactly compensated by the non-5-cycle cross-term. The Arakelov height, being the *sum over all places* (not restricted to 5-cycles), already includes both contributions. Level 3 activates only when the Arakelov sum vanishes ($|N| = 1$ and $b = 0$), leaving the 5-cycle restriction as the sole non-zero contribution. Numerical verification confirms this: naively *adding* the representation cross-term $\pm 2/a_1$ to the Arakelov correction would shift m_t by -30 GeV (105σ) and m_b by -734 MeV (105σ), increasing χ^2 from 10.11 to 2.2×10^4 and categorically excluding the double-counting interpretation.

Significance. The cascade reduces the seven apparently independent correction branches to three levels of a single mechanism: the Galois obstruction on the Wilson

line bundle. The arithmetic filtration $|N| > 1 \supset |N| = 1 \supset b = 0$ is a natural structure of $\mathbb{Z}[\phi]$, not an imposed classification.

Unified coupling. All three cascade levels share the universal coefficient $C = 4/(a_1^2 + 1) = 2/13$. Writing each correction as $\delta(z, \rho) = C \cdot \Omega(z, \rho)$, the three functional forms of Ω are:

Level	$\Omega(z, \rho)$	Origin
1 ($ N > 1$)	$-\ln N(z) /\phi$	Arakelov height
2 ($ N \leq 1, b \neq 0$)	$\phi^3 z' ^{3/4}/4$	Morrey–Sobolev
3 ($b = 0, z \neq 0$)	$-2/a_1 \cdot 1/C$	Character cross-term

The filtration $F^2 \supset F^1 \supset F^0$ (corresponding to Levels 1, 2, 3) is natural from the arithmetic of $\mathbb{Q}(\sqrt{5})$: the Galois information content descends $\mathbb{Z} \rightarrow \mathbb{R} \rightarrow \mathbb{Z}/2$ at each level. However, the three Ω take *irreducible* functional forms (logarithmic, power, constant), and the boundary between Level 2 and Level 3 is discontinuous: the lepton coefficient evaluates to $c_\ell \cdot 1 = +0.163$ while the character cross-term gives $-2/a_1 = -0.400$. The Tate cohomology group $\hat{H}^1(\mathbb{Z}/2, \mathbb{Z}^\times) = \mathbb{Z}/2$ is the obstruction space for Level 3. The three functional forms are partially unified by the Morrey–Sobolev embedding theorem via the effective dimension d_{eff} (Section S5.7); the explicit spectral action computation of normalisation coefficients remains open.

S5.7 Sobolev–Arakelov Interpretation of the Galois Cascade

The three cascade levels (Section S5.6) take irreducible functional forms—logarithmic, power-law, constant—but are unified by the Morrey–Sobolev embedding theorem at varying effective dimension.

Derivation chain. The argument proceeds in six steps:

- (A) *Spectral action $\rightarrow p = 4$ energy.* The spectral action $\text{Tr}(f(D/\Lambda))$ at order Λ^0 contains the Seeley–DeWitt coefficient $a_4 = \text{Tr}(D^4)$. With inner fluctuation $A = z[D_F, z^*]$, this generates $\text{Tr}(A^4) = \sum_{\text{edges}} |A_{ij}|^4$: the $p = 4$ Sobolev energy on the McKay graph. The value $p = d_{\text{ST}} = 4$ is not a choice but a consequence of the spectral action in d_{ST} dimensions.
- (B) *Galois anti-symmetric fluctuation.* The Galois obstruction is carried by $A_- = (A - \sigma(A))/2$. Since $A_{-,ij} = \frac{1}{2}(z_i - z'_i)(D_i - D_j) = \frac{b_i\sqrt{5}}{2}(D_i - D_j)$, the anti-symmetric fluctuation vanishes for rational Wilson lines ($b = 0$), as required for Level 3.
- (C) *Effective dimension from Galois rank.* Define $d_{\text{eff}}(z)$ as the number of independent archimedean degrees of freedom:
 - $b = 0$ (rational): $\sigma(z) = z$, orbit trivial $\Rightarrow d_{\text{eff}} = 0$.
 - $|N(z)| = 1, b \neq 0$ (unit): $|z \cdot z'| = 1$ constrains one combination. By the Dirichlet unit theorem, $\text{rank}(\mathcal{O}_K^\times) = r_1 + r_2 - 1 = 1$ for $K = \mathbb{Q}(\sqrt{5})$, so $d_{\text{eff}} = 1$.
 - $|N(z)| > 1$ (non-unit): both archimedean places contribute independently. The spinorial structure of the Dirac operator provides $d_{\text{ST}}/2 = 2$ effective dimensions per place, giving $d_{\text{eff}} = 2 \times 2 = d_{\text{ST}} = 4$.
- (D) *Morrey–Sobolev embedding.* By the Morrey inequality, $W^{1,p}(\mathbb{R}^d) \hookrightarrow C^{0,\alpha}$ with Hölder exponent $\alpha = 1 - d/p$. At $p = d_{\text{ST}} = 4$:

d_{eff}	$\alpha = 1 - d_{\text{eff}}/4$	Extremal Φ	Cascade level
0	1	constant	Level 3 (d quark)
1	3/4	$ z' ^{3/4}$	Level 2 (leptons)
4	0 (critical)	$\ln N(z) $	Level 1 (c, t, b)

The logarithmic form at $d_{\text{eff}} = p$ (critical Sobolev) and the Hölder form at $d_{\text{eff}} < p$ (subcritical) are both standard results of Morrey–Sobolev theory.

- (E) *Coefficient ϕ^3/d_{ST} .* The Level 2 coefficient $c_\ell = C \cdot \phi^3/d_{\text{ST}}$ involves two quantities: $\phi^3 = 1/(2\alpha_s)$ (the TQFT vacuum energy, derived from $\mathbb{Z}[\phi]$) and $d_{\text{ST}} = \text{Tr}(\phi^3) = \phi^3 - 1/\phi^3$. The latter follows from the algebraic identity $\phi^6 - 1 = 4\phi^3$ (equivalently, $(\phi\phi')^3 = -1$ and $\phi^3 + \phi'^3 = \phi^3 - 1/\phi^3$). Thus:

$$\frac{\phi^3}{d_{\text{ST}}} = \frac{\phi^3}{\phi^3 - 1/\phi^3} = \frac{\phi^6}{\phi^6 - 1} \quad (\text{S11})$$

is a derived quantity, not an independent parameter.

- (F) *Signs.* The sign within each level is determined by the Galois action: $\text{sign}(z')$ for leptons (Morrey extremal with signed argument); $\sigma(\phi) = -1/\phi$ for the b quark (Galois conjugation of the Wilson line); arm topology on the McKay graph determines the sign of spectral corrections (Section S9.1).

Summary. The full derivation chain is:

$$\underbrace{\text{Spectral action}}_{\text{(standard NCG)}} \rightarrow \underbrace{\text{Tr}(D^4)}_{\text{(identity)}} \rightarrow \underbrace{\|A_-\|_4^4}_{\text{(Galois)}} \rightarrow \underbrace{\text{Morrey at } d_{\text{eff}}}_{\text{(mathematics)}} \rightarrow \underbrace{\alpha = 1 - d_{\text{eff}}/4}_{\text{(Hölder)}}$$

Each step is either standard noncommutative geometry, standard Sobolev theory, or derived from the 600-cell framework.

Remaining gaps. (i) The $d_{\text{eff}} = d_{\text{ST}}$ assignment for non-units relies on the spinorial doubling argument (two effective dimensions per archimedean place), which is structurally motivated but not rigorously proved. (ii) The Level 2→Level 3 boundary remains discontinuous ($c_\ell \cdot 1 = +0.163$ vs. $-2/a_1 = -0.400$); this is structural, not a deficiency—different d_{eff} regimes have different extremals. (iii) The explicit computation of the Morrey normalisation from the spectral action coupling constants has not been carried out. The unification is therefore PARTIAL: the three functional forms are identified as regimes of a single embedding, but the full variational principle on the arithmetic McKay space $X = \text{Spec}(\mathbb{Z}[\phi]) \times_{\mathbb{Z}/2} \text{McKay}(2I)$ remains conjectural.

S5.8 Galois Norm Sum Rules: Proof and Consequences

This section provides the full proof of Theorem ?? (Section ?? of the main text) and derives its consequences.

Galois concordance. For a Wilson line $z = a + b\phi \in \mathbb{Z}[\phi]$, define z as *concordant* if z and $\sigma(z) = a + b\phi'$ have the same sign ($N(z) = z\sigma(z) > 0$), and *discordant* if they have opposite signs ($N(z) < 0$). The concordance pattern of the nine fermions is:

Fermion	(a, b)	z	$\sigma(z)$	$N(z)$	Level	Concordance
e	$(0, 0)$	0	0	0	0	degenerate
μ	$(1, 1)$	2.618	0.382	+1	2	concordant
τ	$(1, 2)$	4.236	-0.236	-1	2	discordant
u	$(3, -2)$	-0.236	4.236	-1	2	discordant
d	$(1, 0)$	1	1	+1	3	concordant
s	$(1, 1)$	2.618	0.382	+1	2	concordant
c	$(2, 1)$	3.618	1.382	+5	1	concordant
b	$(-1, 4)$	5.472	-3.472	-19	1	discordant
t	$(4, 1)$	5.618	3.382	+19	1	concordant

Full proof of Theorem ??. The proof proceeds by cascade level.

Level 2 cancellation. At Level 2, all fermions have $|N(z)| = 1$, so $N = \pm 1$. The pairs (μ, τ) and (s, u) each contain one concordant member ($N = +1$) and one discordant member ($N = -1$). This pairing is structural: the cascade’s anti-symmetric sector requires that within each pair, exactly one member crosses the “Galois wall” ($\sigma(z)$ changes sign relative to z , reflecting the two real embeddings of $\mathbb{Q}(\sqrt{5})$). Therefore $\sum_{\text{Level } 2} N(z_f) = 0$.

Level 1 cancellation. For t and b :

1. $N(z_t) \cdot N(z_b) = N(z_t z_b) = N(19\phi) = 19^2 \cdot N(\phi) = -361 < 0$. Therefore the norms have *opposite signs*.
2. $|N(z_t)| \cdot |N(z_b)| = 361 = 19^2$. The integer factorisations of 361 are 1×361 and 19×19 .
3. The irreducibility condition (C1) of Theorem ?? requires $|N(z_f)|$ to be 0, 1, or a prime that splits or ramifies in $\mathbb{Q}(\sqrt{5})$. Since $361 = 19^2$ is *not* prime, $|N| = 361$ is excluded.
4. Therefore $|N(z_t)| = |N(z_b)| = 19$, and combined with opposite signs, $N(z_t) + N(z_b) = 0$.

The charm quark is unpaired: $N(z_c) = N(2 + \phi) = 4 + 2 - 1 = 5 = a_1$.

Charm norm identity. The charm Wilson line $z_c = 2 + \phi = (a_1 + \sqrt{a_1})/2$ has norm $N(z_c) = a_1(a_1 - 1)/4$, which equals a_1 if and only if $a_1 = 5$. This self-consistency condition is a consequence of, not an input to, the framework.

Levels 0 and 3. $N(z_e) = N(0) = 0$ and $N(z_d) = N(1) = 1$.

Total. $\sum N = 0 + (a_1 + 0) + 0 + 1 = a_1 + 1 = b_1$. □

Component sums. The cascade concordance structure implies:

$$\begin{aligned} \sum_f a_f &= 12 = \dim(G_{\text{SM}}), & \sum_f b_f &= 8 = \text{rank}(E_8), \\ \sum_f a_f b_f &= 0 \quad (\text{orthogonality}), & \sum_f n_f &= 108 = 9 \times 12 = N_{\text{eig}} \cdot \dim(G). \end{aligned} \tag{S12}$$

These are *consequences* of the cascade pairing, not independent constraints.

Cubic identity. The same concordance mechanism gives $\sum N^3 = a_1^3 + 1 = 126 = |2I| + b_1$. The equation $a_1^3 + 1 = a_1! + a_1 + 1$ reduces to $(a_1 - 2)! = a_1 + 1$, which is the secondary Diophantine (??) re-derived from Galois norms. This constitutes a twelfth independent characterisation of $a_1 = 5$.

Quadratic identity. Unlike odd powers, even powers do not benefit from concordance cancellation ($19^2 + (-19)^2 = 722 \neq 0$). The quadratic sum is $\sum N^2 = 752 = \text{rank}(E_8) \cdot (|2I| - (a_1^2 + 1)) = 8 \times 94$.

TQFT vacuum connection. The TQFT vacuum energy on the A_3 fusion ring (SU(2) at level $k = a_1 - 2 = 3$) is

$$E_{\text{vac}} = \frac{1}{2} \sum_{j>0} d_j \lambda_j = \phi^3 = \frac{1}{2\alpha_s} \quad (\text{S13})$$

connecting the Verlinde ground-state energy to the inverse strong coupling. The Galois conjugate $E'_{\text{vac}} = \phi'^3 = 2 - \sqrt{5} < 0$ gives $E_{\text{vac}} \cdot E'_{\text{vac}} = -1$ (a unit norm), and the neutrino mass ratio becomes $r_\nu = \Delta m_{21}^2 / \Delta m_{31}^2 = \alpha \cdot E_{\text{vac}} = \alpha / (2\alpha_s)$ —the ratio of electromagnetic to strong coupling constants.

Leptonic norm vanishing. $\sum_{\text{leptons}} N(z_f) = N(z_e) + N(z_\mu) + N(z_\tau) = 0 + 1 + (-1) = 0$. This vanishing is analogous to gravitational anomaly cancellation and is a direct consequence of the concordant/discordant pairing in the lepton sector.

Robustness. Among all 11,664 alternative (a, b) selections with $|N| \leq 30$ for the same exponents $\{n_f\}$, only 86 (0.7%) yield $\sum N = 6$. The $Z[\phi]$ selection (C1 + C2) picks the unique physical solution from 5.4×10^8 candidates, and the norm sum rule holds for this unique choice.

A thirteenth characterisation of $a_1 = 5$ (CKM concordance) is given in Section S9.1; a fourteenth (Cayley–CKM vacuum offset) in Section S9.2.

S5.9 Doublet Norm Products

The product of up-type and down-type Wilson-line norms within each generation reveals a striking pattern:

Generation	Up/Down	$z_{\text{up}} \cdot z_{\text{down}}$	$ N(z_{\text{up}} \cdot z_{\text{down}}) $	V
1	u, d	$(3 - 2\phi)(1) = 3 - 2\phi$	$ 9 - 12 + 4 = 1$	1
2	c, s	$(2 + \phi)(1 + \phi) = 2 + 3\phi + \phi^2 = 3 + 4\phi$	$ 9 + 12 - 16 = 5$	a_1
3	t, b	$(4 + \phi)(-1 + 4\phi) = -4 + 15\phi + 4\phi^2 = 0 + 19\phi$	$ 0 + 0 - 361 = 361$	(4)

The sequence 1, $a_1 = 5$, $(4a_1 - 1)^2 = 361$ is a monotonically increasing function of a_1 alone. The full 3×3 norm matrix $M_{ij} = |N(z_{u_i} \cdot z_{d_j})|$ is rank-1, indicating that the doublet products are controlled by a single degree of freedom.

The sum $N(z_{\text{up}}) + N(z_{\text{down}})$ by generation gives $(-1) + (+1) = 0$ (Gen 1), $(+5) + (+1) = 6$ (Gen 2), $(+19) + (-19) = 0$ (Gen 3): only the second generation contributes. The Planck Wilson line $z_P = 4\phi^2 = 4(\phi + 1) = 4 + 4\phi$ has $N(z_P) = 16 - 16 = 16 + 16 - 16 = \text{Tr}(A^2) = p_2 = (a_1 - 1)^2$, connecting the hierarchy to the second McKay power sum.

Assessment. The scaling $1 \rightarrow a_1 \rightarrow (4a_1 - 1)^2$ is verified numerically but lacks a derivation from dynamics. No CKM connection was found (the rank-1 structure is too degenerate). Classification: PATTERN.

S5.10 The Koide Mass Relation

The Koide formula [2] states that for charged leptons,

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (\text{S14})$$

Experimentally $Q = 0.666661 \pm 0.000007$, matching $2/3$ to 0.001%.

Framework prediction. With the corrected masses, the framework gives $Q = 0.66668$, reproducing the relation to 0.002%. The bare exponents give $Q_{\text{bare}} = 0.6728$, off by 0.92%. The Morrey corrections shift Q toward $2/3$ by a factor of 400.

Palindromic connection to $a_1 = 5$. For a geometric sequence $m_i = m_0 r^i$, the condition $Q = 2/3$ reduces to the palindromic quartic $s^4 - 4s^3 - 3s^2 - 4s + 1 = 0$. Substituting $t = s + 1/s$ collapses this to $(t - 5)(t + 1) = 0$, giving $t = a_1$ as the unique positive solution.

Assessment. The Koide relation is *not* an exact algebraic identity in the framework. Nevertheless, the framework reproduces $Q = 2/3$ to 0.002% with zero free parameters. Classification: PATTERN.

S5.11 One-Loop QED Coefficient: Verlinde Derivation

This section derives the one-loop QED coefficient $c_{\text{QED}} = \sqrt{2/a_1}$ appearing in the one-loop mass formula of the main text (Section 8.7). The derivation uses only the modular S -matrix of $SU(2)_k$ Chern–Simons theory at level $k = a_1 - 2 = 3$.

Setup. The $SU(2)_k$ fusion ring at level $k = 3$ has $k + 1 = 4$ simple objects (spins $j = 0, \frac{1}{2}, 1, \frac{3}{2}$) with quantum dimensions

$$d_j = \frac{\sin((2j+1)\pi/r)}{\sin(\pi/r)}, \quad r = k + 2 = a_1 = 5 \quad (\text{S15})$$

giving $d_0 = 1$, $d_{1/2} = \phi$, $d_1 = \phi$, $d_{3/2} = 1$. The total quantum dimension is $\mathcal{D}^2 = \sum_j d_j^2 = a_1 + \sqrt{a_1}$. The modular S -matrix has entries

$$S_{jl} = \sqrt{\frac{2}{r}} \sin\left(\frac{(2j+1)(2l+1)\pi}{r}\right) \quad (\text{S16})$$

satisfying unitarity $SS^\dagger = \mathbf{1}$ and the Verlinde formula

$$N_{ij}^k = \sum_l \frac{S_{il} S_{jl} \overline{S_{kl}}}{S_{0l}} \quad (\text{S17})$$

Self-energy on the fusion ring. The one-loop self-energy of a particle of spin j propagating on the quantum McKay graph (the fusion graph of the fundamental representation $j = \frac{1}{2}$) involves the fusion vertex $j \otimes \frac{1}{2} \rightarrow k$ and the propagator (quantum Green's function). Using the Verlinde formula to decompose the fusion coefficients, the self-energy contains the factor

$$F(j, l) = \frac{S_{jl} S_{\frac{1}{2}, l}}{S_{0l}} = \sqrt{\frac{2}{r}} \sin\left(\frac{(2j+1)(2l+1)\pi}{r}\right) \cdot \lambda_l \quad (\text{S18})$$

where $\lambda_l = S_{\frac{1}{2}, l}/S_{0l} = 2 \cos((2l+1)\pi/r)$ is the eigenvalue of the quantum adjacency matrix $N_{1/2}$.

Counting $\sqrt{2/r}$ factors. The key observation is:

- The numerator S_{jl} contributes one factor of $\sqrt{2/r}$ from Eq. (S16).
- The numerator $S_{\frac{1}{2}, l}$ contributes one factor of $\sqrt{2/r}$.
- The denominator $1/S_{0l}$ contributes $1/\sqrt{2/r} = \sqrt{r/2}$.

The net factor is $(\sqrt{2/r})^2/\sqrt{2/r} = \sqrt{2/r} = \sqrt{2/a_1}$.

The remaining factors—the sine function $\sin((2j+1)(2l+1)\pi/r)$ and the eigenvalue λ_l —depend on j and l . After summing over the internal label l and contracting with the Green’s function, these produce the j -dependent self-energy $\Sigma(j)$. But the universal prefactor $\sqrt{2/r}$ factors out of the sum:

$$\Sigma(j) = \sqrt{\frac{2}{r}} \cdot \Sigma_{\text{red}}(j) \quad (\text{S19})$$

where $\Sigma_{\text{red}}(j)$ is the reduced self-energy that varies with j .

Identification of c_{QED} . In the mass formula, the one-loop correction is $\delta_2 = c_{\text{QED}} \alpha \delta_1$. The coefficient c_{QED} must be *universal* (the same for all fermions), since α is a universal coupling and δ_1 already carries the j -dependent information. The only j -independent factor in the self-energy is $\sqrt{2/r}$; the j -dependent part $\Sigma_{\text{red}}(j)$ is absorbed into δ_1 . Therefore:

$$c_{\text{QED}} = \sqrt{\frac{2}{r}} = \sqrt{\frac{2}{a_1}} = \sqrt{\frac{2}{5}} = 0.63246 \quad (\text{S20})$$

Three arguments for universality.

1. *S-matrix structure:* The factor $\sqrt{2/r}$ is the normalisation of the S -matrix (S16), determined by unitarity $SS^\dagger = \mathbf{1}$. It depends only on $r = a_1$, not on any spin label.
2. *Perturbative separation:* In standard QFT renormalisation, the vertex factor (from the S -matrix) and the propagator (from the Green’s function) contribute independently. The universal coefficient c_{QED} is a vertex factor; the j -dependent part resides in the propagator.
3. *Numerical exclusion:* Using $c = \Sigma(\frac{1}{2}) = 0.409$ instead of $\sqrt{2/5}$ gives a muon pull of -2851σ —excluded by $> 1000\sigma$. Only $c = \sqrt{2/a_1}$ is consistent with data.

Numerical verification. The S -matrix unitarity, Verlinde algebra, and fusion rules are verified to 10^{-16} precision. The predicted coefficient $c_{\text{QED}} = 0.63246$ matches the value required by the global χ^2 fit ($c_{\text{needed}} = 0.63225$) to 0.032%—a 2.57σ pull driven entirely by the muon, whose mass is known to 6 ppb.

Classification: DERIVED.

S5.12 Statistical Assessment

This section provides the full error analysis for the fermion mass predictions.

Pull table. The complete results including one-loop corrections are shown in Table S2.

Fermion	(a, b)	m_{pred}	$m_{\text{PDG 2024}}$	Pull (σ)	Scheme
e	(0, 0)	0.5110 MeV	0.5110 MeV	input	pole
μ	(1, 1)	105.6584 MeV	105.6584 MeV	+2.57	pole
τ	(1, 2)	1776.75 MeV	1776.86 MeV	-0.96	pole
u	(3, -2)	2.165 MeV	2.16 MeV	+0.07	$\overline{\text{MS}}(2 \text{ GeV})$
d	(1, 0)	4.671 MeV	4.70 MeV	-0.42	$\overline{\text{MS}}(2 \text{ GeV})$
s	(1, 1)	93.38 MeV	93.5 MeV	-0.15	$\overline{\text{MS}}(2 \text{ GeV})$
c	(2, 1)	1271 MeV	1273 MeV	-0.30	$\overline{\text{MS}}(m_c)$
b	(-1, 4)	4173 MeV	4183 MeV	-1.49	$\overline{\text{MS}}(m_b)$
t	(4, 1)	172.67 GeV	172.57 GeV	+0.33	pole
$\chi^2 = 10.11 / 8 \text{ dof}$				$p = 0.26, \text{ RMS} = 1.12 \sigma$	

Table S2: Complete pull table for fermion masses with one-loop QED correction. PDG 2024 values; mixed mass scheme.

Discriminating vs. non-discriminating tests. Of the eight predictions, only two— μ and τ —have experimental uncertainties small enough to produce pulls $> 1\sigma$. The six quark masses have uncertainties of 1–7 MeV (relative 0.2–3%), far larger than the framework’s sub-percent deviations. Thus μ and τ are the *discriminating* tests; the quarks confirm consistency but cannot distinguish this framework from other models with comparable precision.

The muon pull (+2.57 σ). The muon mass is measured to 6 ppb, making it the most stringent test. The +2.57 σ pull traces directly to the Morrey exponent: the tree-level correction $\delta_1(\mu) = c_\ell \cdot |z'_\mu|^{3/4}$ depends on $k = 3/4$ through the 3/4-th power of $|z'_\mu| = 1/\phi^2$. A shift $k \rightarrow 0.7496$ (vs. $3/4 = 0.7500$) would eliminate the pull entirely.

The pull is *irreducible* within the framework. Two natural corrections have been investigated and both fail to reduce it: (i) the two-loop self-energy on the $SU(2)_3$ fusion ring (dressed propagator plus vertex correction) is positive—the same sign as the one-loop—and therefore *increases* the pull; (ii) discrete Morrey corrections on the finite McKay graph do not modify the Hölder exponent, which is determined by the ambient spacetime dimension $d_{\text{ST}} = 4$, not by the graph geometry. A coefficient renormalisation $c_\ell \rightarrow c_\ell(1 + \sqrt{2/a_1} \alpha)$ is also ruled out: it shifts the muon prediction in the wrong direction. The +2.57 σ pull is therefore a genuine prediction of $k = 3/4$ exactly, and the data are consistent at this level.

The bottom quark pull (-1.49 σ). The second-largest pull comes from b , driven by the $\overline{\text{MS}}(m_b)$ mass uncertainty of 7 MeV. The prediction 4173 MeV vs. 4183 ± 7 MeV is a 1.49 σ deficit, consistent with statistical fluctuation.

Scheme dependence. Using pole masses for all quarks (instead of the mixed scheme) would shift charm and bottom pulls significantly, since the pole-to- $\overline{\text{MS}}$ conversion for these quarks is $\sim 10\%$. The mixed scheme is physically motivated: the geometric formula computes UV (short-distance) masses, matching the $\overline{\text{MS}}$ definition. Pole masses are used only where the two definitions coincide at the framework’s precision (leptons and top).

Overall assessment. A χ^2 of 10.11 for 8 degrees of freedom corresponds to $p = 0.26$ —a perfectly acceptable fit. The framework has zero adjustable parameters beyond m_e , so there is no overfitting risk. The RMS pull of 1.12 σ is close to the ideal value of 1.0 σ expected for a correct model with Gaussian errors.

Classification: DERIVED (all corrections); PATTERN (overall fit quality).

S6 Electroweak: Spectral Weight Analysis

This expands on Section 10 of the main text.

S6.1 The Spectral Weight

For each irrep ρ_k of $2I$ with dimension d_k and character χ_k at the nearest-neighbor conjugacy class, define the *spectral weight*:

$$z_k = \frac{\text{degree} \cdot \chi_k^2}{d_k} = \frac{12 \chi_k^2}{d_k} \quad (\text{S21})$$

Irrep	d_k	$\chi_k(12A)$	z_k	$N(z_k)$	$\sqrt{N(z_k)}$
R_1	1	1	12	144	12
R_2	2	ϕ	$6\phi^2 = 6 + 6\phi$	36	6
R'_2	2	$-1/\phi$	$6/\phi^2 = 6 - 6\phi'$	36	6
R_3	3	ϕ	$4\phi^2 = 4 + 4\phi$	16	4
R'_3	3	$-1/\phi$	$4/\phi^2$	16	4
R_4	4	1	3	9	3
R'_4	4	-1	3	9	3
R_5	5	0	0	0	0
R_6	6	-1	2	4	2

The Planck exponent is the spectral weight of R_3 :

$$z_{\text{Planck}} = z_{R_3} = \frac{12\phi^2}{3} = 4\phi^2 \quad (\text{S22})$$

The electromagnetic–gravitational factorization. The linear coefficient of the α equation factorizes as:

$$B = 4a_1\phi^4 = a_1 \cdot \phi^2 \cdot z_{\text{Planck}} \quad (\text{S23})$$

Self-consistency and uniqueness. The norm condition $N(z_{\text{Planck}}) = (a_1 - 1)^2$ requires $a_1 - 1 = 2(a_1 + 1)/3$, which has the *unique* solution $a_1 = 5$.

Mass formula coefficients from the spectral weight.

$$a_1 = \sqrt{N(z_{\text{Planck}})} + 1 = \sqrt{16} + 1 = 5 \quad (\text{S24})$$

$$b_1 = \text{Tr}(z_{\text{Planck}})/2 = 12/2 = 6 \quad (\text{S25})$$

Total spectral weight. Direct computation gives:

$$\sum_{k=1}^9 z_k = 2a_1^2 = 50 \quad (\text{S26})$$

S7 Gravity: Extended Results

This expands on Section 12 of the main text.

S7.1 Graviton Spectrum and Gap-Planck Identities

The coexact Laplacian C has 21 distinct nonzero eigenvalues. The **coexact spectral gap** is:

$$\lambda_1^{\text{coexact}} = 7 - 4\phi, \quad N(7 - 4\phi) = a_1 = 5 \quad (\text{S27})$$

The spectral gaps relate to the Planck exponent through product identities:

$$\lambda_1^{\text{exact}} \times z_{\text{Planck}} = (12 - 6\phi)(4 + 4\phi) = 24 = |2T| \quad (\text{S28})$$

$$\lambda_1^{\text{coexact}} \times z_{\text{Planck}} = (7 - 4\phi)(4 + 4\phi) = 12 - 4\phi = \lambda_2(\Delta_0) \quad (\text{S29})$$

S7.2 Galois Kernel Theorem: Full Proof

Theorem S2 (Galois Kernel Theorem). *On the 600-cell with any Hopf fibration:*

$$\ker(\square) = \ker(b_1 A_{\text{fiber}} - A) = \rho_0 \oplus \rho_1 \oplus \rho_8 \quad (\text{S30})$$

with $\dim(\ker) = 1 + 4 + 4 = 9 = N_{\text{eig}}$.

Proof. The kernel condition $Av = b_1 A_{\text{fiber}} v$ requires v to be a simultaneous eigenvector of both operators, with eigenvalues satisfying $\lambda_A = b_1 \mu_{\text{fiber}}$. Each Hopf fiber is a C_{10} cycle (10 vertices), whose eigenvalues are $2 \cos(k\pi/a_1)$ for $k = 0, 1, \dots, a_1 - 1$. The 600-cell adjacency eigenvalues divided by b_1 must match a C_{10} eigenvalue:

Irrep	λ_A	λ_A/b_1	C_{10} eigenvalue?	Multiplicity
ρ_0 (1)	12	2	$2 \cos(0) = 2$	Yes (1)
ρ_1 (2)	6ϕ	ϕ	$2 \cos(\pi/5) = \phi$	Yes (4)
ρ_8 (2')	$6\phi'$	ϕ'	$2 \cos(3\pi/5) = \phi'$	Yes (4)
ρ_2 (3)	4ϕ	$2\phi/3$	—	No
ρ_3 (4)	3	$1/2$	—	No
ρ_4 (5)	0	0	—	No

The fiber angular momenta of the matching modes are $k = 0, 1, 3$; mode $k = 2$ is absent because $b_1/\phi = 6/\phi \notin \text{spec}(A)$. Numerical verification confirms $\dim(\ker) = 9$ for all 27 distinct Hopf fibrations of the 600-cell. $\square$

Proposition S2 (Uniqueness of b_1). *Among all integers $c = 1, \dots, 2a_1+2$, only $c = b_1 = 6$ gives a nontrivial kernel with $\dim > 1$.*

Galois structure. The kernel selects the *smallest* irrep from each Galois orbit of $2I$:

- Rational orbit: selects ρ_0 (dim 1)
- ϕ -orbit: selects ρ_1 (dim 2)
- ϕ' -orbit: selects ρ_8 (dim 2')

The sum of kernel irrep dimensions is $1 + 2 + 2 = 5 = a_1$.

S7.3 Spectral Balance, Spectral Sum Rules, and Propinquity Bound

Proposition S3 (Spectral balance). *The simplicial Dirac operator $D = d + d^*$ on the 600-cell satisfies*

$$\text{Str}_{\text{form}}(D^{2k}) \equiv \text{Tr}_{\text{even}}(D^{2k}) - \text{Tr}_{\text{odd}}(D^{2k}) = 0 \quad \forall k \geq 0, \quad (\text{S31})$$

where the grading is by form-degree parity: even forms ($\Omega^0 \oplus \Omega^2$, dimension 1320) versus odd forms ($\Omega^1 \oplus \Omega^3$, dimension 1320).

Spectral sum rules. The coexact Laplacian $C = d_1^T d_1$ has a uniform diagonal equal to $a_1 = 5$, giving $\text{Tr}(C) = a_1 \cdot E = 5 \times 720 = 3600$. The coexact spectrum exhibits Galois pairing.

Hopf fibration connection. The 720 edges of the 600-cell are partitioned by $b_1 = 6$ Hopf fibrations, each consisting of 12 great-circle fibers of 10 edges (decagons): $E = b_1 \times 12 \times 10 = 6 \times 120 = 720$.

Propinquity bound. The Cayley graph of $2I$ approximates S^3 with a Latrémolière propinquity bound:

$$\Lambda^* \leq \varepsilon = 0.386 \approx \frac{\pi}{a_1 \phi} \quad (12.3\% \text{ of } \text{diam}(S^3)). \quad (\text{S32})$$

S7.4 Emergent Graviton: Polarization Ratio

We show that the 600-cell Regge calculus produces edge perturbations with exactly the correct polarization content for spin-2 gravity, despite the absence of a fundamental graviton mode.

No fundamental graviton. The coexact Laplacian $C = d_1^T d_1$ has 21 distinct nonzero eigenvalues with multiplicities:

$$\{6, 16, 30, 48, 30, 40, 36, 24, 16, 24, 72, 40, 30, 16, 25, 48, 24, 16, 30, 24, 6\} \quad (\text{S33})$$

summing to 601. The multiplicity 9—the dimension of $\text{Sym}_0^2(\mathbb{R}^4)$, the graviton representation under the 600-cell symmetry group H_4 —is *absent* from the coexact (physical) spectrum. It does appear in the *exact* (gauge) sector at eigenvalues $12 - 4\phi$ and $8 + 4\phi$, confirming that the H_4 representation exists but is “consumed” by gauge degrees of freedom (discrete diffeomorphisms).

Level assignment. On the round S^3 , the scalar harmonics at level l have multiplicity $(l+1)^2$, while the coexact 1-form harmonics have multiplicity $2l(l+2)$. The 600-cell graph Laplacian Δ_0 has scalar multiplicities $\{1, 4, 9, 16, 25, 36, 9, 16, 4\}$: the first six match $(l+1)^2$ for $l = 0, \dots, 5$ exactly (91 modes), while the remaining $\{9, 16, 4\}$ are lattice-aliased modes (Brillouin zone folding).

For the coexact sector, the first four eigenvalues have multiplicities $\{6, 16, 30, 48\}$, matching $2l(l+2)$ for $l = 1, 2, 3, 4$ as single eigenspaces. At $l = 5$, the continuum multiplicity $70 = 2 \times 5 \times 7$ splits across two discrete eigenvalues ($30 + 40 = 70$), preserving the total. Beyond $l = 5$, lattice artifacts prevent clean level assignment.

The polarization ratio. For each matched level $l = 1, \dots, 5$:

l	$\text{mult}_{\text{scalar}}$	$\text{mult}_{\text{coexact}}$	$R(l)$	Value
1	4	6	$6/4 = 3/2$	1.5000
2	9	16	$16/9$	1.7778
3	16	30	$30/16 = 15/8$	1.8750
4	25	48	$48/25$	1.9200
5	36	70	$70/36 = 35/18$	1.9444

All five ratios are *exact* (integer multiplicities). The algebraic identity $R(l) = 2 - 2/(l+1)^2$ converges to $R = 2$ as $l \rightarrow \infty$.

Physical interpretation: 2 = graviton polarizations. In 3+1-dimensional General Relativity, the metric perturbation $h_{\mu\nu}$ has 10 components; diffeomorphism gauge removes 4, the trace (conformal mode) removes 1, and the Hamiltonian and momentum constraints remove 3, leaving exactly $10 - 4 - 1 - 3 = 2$ physical polarizations (helicities ± 2). On the spatial slice S^3 , these 2 polarizations appear at each angular momentum level as the ratio of transverse-traceless tensor modes to scalar modes.

Three properties establish the emergent graviton:

1. **TT projection is exact.** The coexact projector $P_{\text{coexact}} = P_{TT}$: $d_0^T \cdot G = 0$ to machine precision (10^{-14}). In Regge calculus, coexact = divergence-free = transverse; the trace mode is removed by the Hodge decomposition.
2. **Correct polarization count.** $R(l) = 2l(l+2)/(l+1)^2$ matches the continuum S^3 at all five resolved levels with zero deviation. This is a *topological* property: it depends only on integer multiplicities, not on eigenvalues or lattice spacing.
3. **Ghost-free propagator.** The Green's function $G = (C|_{\text{coexact}})^{-1}$ is positive-definite (all 601 eigenvalues positive), with spectral gap $\lambda_1 = 7 - 4\phi > 0$.

Galois structure. The coexact eigenvalues come in Galois-conjugate pairs with identical multiplicities: $(7 - 4\phi, 3 + 4\phi)$ both with multiplicity 6; $(6 - 3\phi, 3 + 3\phi)$ both with multiplicity 16; etc. The Galois pairing preserves the polarization ratio at each level, ensuring that the dark-sector coexact spectrum has the same graviton content—consistent with the universal gravitational coupling derived in Section S7.

Dispersion relation. The coexact eigenvalues are not exactly proportional to $l(l+2)$: the ratio $\lambda_{\text{coexact}}(l)/l(l+2)$ varies by 18.6% across $l = 1, \dots, 5$. However, the scalar eigenvalues show the *same* distortion (22.0% variation), and the ratio $\lambda_{\text{coexact}}(l)/\lambda_{\text{scalar}}(l)$ is approximately constant at 0.23 ± 0.02 (7.9% variation). A power-law fit gives $\lambda \propto [l(l+2)]^p$ with $p = 0.79$ (coexact) and $p = 0.75$ (scalar)—both deviate from $p = 1$ by the same amount. This is a universal lattice artifact: on any finite triangulation of S^3 , eigenvalues are compressed relative to the continuum, particularly at high angular momentum (analogous to $\sin^2(k/2)$ versus k^2 on a cubic lattice). The key physical content is that tensor and scalar modes see the same effective geometry, consistent with a massless graviton in the continuum limit.

Conclusion. The graviton does not appear as a single fundamental mode (multiplicity 9 is absent from the physical sector). Instead, it *emerges* from the collective coexact spectrum: at each angular momentum level, the edge perturbations carry exactly two transverse polarizations, reproducing the representation-theoretic content of linearized General Relativity. The emergence is *exact* at resolved levels, not an asymptotic approximation. The dispersion relation is consistent with massless propagation in the continuum limit, with deviations attributable to universal lattice effects.

S7.5 E_8 Coxeter Decompositions and the Grassmannian Angle

The Coxeter projection $\mathbb{R}^8 \rightarrow \mathbb{R}^4$ (onto the H_4 eigenspace with exponents $\{1, 11, 19, 29\}$) maps the 240 E_8 roots to two concentric 600-cells of projected norms $n^2 = 1 \pm 1/\sqrt{a_1}$, giving norm ratio ϕ . Exhaustive enumeration of all $8!$ orderings of simple reflections yields exactly

$$40 = \frac{N}{b_1} = \text{rank}(E_8) \times a_1 \quad (\text{S34})$$

distinct decompositions of the 240 roots into inner/outer sets. All 40 are isomorphic as E_8 root subsets (identical Gram spectra and adjacency structure). The identity $40 = h/b_1 \times \text{rank}$ follows from $h(E_8) = a_1 b_1$ and the stabiliser order $|\text{Stab}| = b_1 \cdot |W(E_7)|$, giving $40 = |W(E_8)|/|\text{Stab}|$.

Each decomposition defines a 4-plane $V \in \text{Gr}(4, 8)$. For all $\binom{40}{2} = 780$ pairs, the combinatorial overlap (number of shared inner roots) satisfies an *exact* Grassmannian formula:

$$\text{overlap}(V_i, V_j) = h(E_8) \sum_{k=1}^4 \cos^2 \theta_k \quad (\text{S35})$$

where $\theta_1, \dots, \theta_4$ are the principal angles between V_i and V_j . The sum of squared cosines takes exactly four values $2m/a_1$ for $m = a_1, \dots, \text{rank}(E_8)$, giving overlaps 60, 72, 84, 96 (always multiples of the vertex degree 12). The number of overlap levels equals $\text{rank} - a_1 + 1 = 4 = \dim V$.

Among these, the 132 pairs whose H_4 subspaces share a two-dimensional Coxeter plane (the eigenspace of the lowest exponent $m = 1$) have singular values $(1, 1, \sqrt{2/a_1}, \sqrt{2/a_1})$. This follows from the overlap formula and conjugate symmetry of exponents $\{11, 19\}$: two shared directions contribute $\cos^2 = 1$ each; the remaining two conjugate eigenplanes carry equal angles, forcing each $\cos^2 = 2/a_1$.

The principal angle $\cos \theta = \sqrt{2/a_1}$ coincides with the bare mass-correction coefficient c_{bare} of Eq. (??). Both values trace to the Plancherel measure $2/a_1$ of the Verlinde ring $\mathbb{Z}[\phi]$ at level $k+2 = a_1$: the S -matrix prefactor $\sqrt{2/a_1}$ normalises the modular transform, and the Grassmannian angle quantises overlap in the same units. The Grassmannian interpretation provides a geometric reason why the mass formula involves $\sqrt{2/a_1}$ (an amplitude) rather than $2/a_1$ (a probability): principal angles are inherently amplitude-level quantities.

A comparison test with $D_6 \rightarrow H_3$ (the only other non-crystallographic Coxeter projection in $\text{rank} \leq 8$) fails: the doubled exponent $m = 5$ in D_6 produces five projected-norm shells instead of two. The clean $120+120$ split and its Grassmannian structure are *specific to* E_8 .

S8 Neutrinos: Exponent Decompositions and Baryogenesis

This expands on Section 11 of the main text.

S8.1 Eight Decompositions of 35

The integer 35 admits at least eight independent expressions in framework constants:

Identity	Value	Interpretation
$b_1^2 - 1$	$36 - 1$	$\dim(\text{ad } SU(b_1))$
$h(E_8) + a_1$	$30 + 5$	Coxeter number + diameter
$a_1(a_1 + 2)$	5×7	quadratic Casimir numerator, $\text{spin-}a_1/2$
$\dim(\mathcal{H}_F) + \text{diam}(\Gamma)$	$30 + 5$	fiber Hilbert space + graph diameter
$n_{\text{top}} + N_{\text{eig}}$	$26 + 9$	heaviest fermion exponent + irrep count
$\sum_e w_e + b_1$	$29 + 6$	total McKay weight + vertex half-degree
$L(\rho_1) \cdot L(\rho_8) - 1$	$b_1^2 - 1$	Galois norm of kernel Laplacians
$\binom{b_1+1}{N_{\text{gen}}}$	$\binom{7}{3}$	combinatorial
$ \eta_A = N/2 - a_1^2$	$60 - 25$	spectral asymmetry of Cayley graph

S8.2 Baryogenesis

The baryon asymmetry η_B can be estimated from framework quantities:

$$\eta_B \sim \alpha_W^4 \cdot J \cdot |\eta| \approx \left(\frac{0.00730}{6/26} \right)^4 \times 3.12 \times 10^{-5} \times 35 \approx 1.1 \times 10^{-9} \quad (\text{S36})$$

The observed value $\eta_B^{\text{obs}} = 6.1 \times 10^{-10}$ differs by a factor ~ 1.8 . All inputs are derived from $a_1 = 5$.

S9 Mixing Angles: CKM Correction Coefficients

This expands on Section 9 of the main text.

The perturbative correction coefficients $\{2/3, 5, 6/26\}$ are ratios of the E_8 Dynkin leg dimensions:

Angle	Correction	Value	E_8 origin	Physics
θ_{12}	$1 - \frac{2\alpha_s}{3\pi}$	$c_{12} = \frac{2}{3} = \frac{b_1}{N_{\text{eig}}}$	Leg B/ N_{eig}	$C_F(SU(3))/2$
θ_{23}	$1 + \frac{5\alpha_s}{\pi}$	$c_{23} = 5 = a_1$	Leg A – Leg B	FN propagation
θ_{13}	$1 + \sin^2 \theta_W$	$c_{13} = \frac{6}{26} = \sin^2 \theta_W$	Leg B/ $(a_1^2 + 1)$	EW correction

Derivation of c_{12} (McKay graph). On the McKay graph $\hat{E}_8$, the irrep ρ_1 has $\dim(\rho_1) = 2$ and ρ_2 has $\dim(\rho_2) = 3$. Their ratio $\dim(\rho_1)/\dim(\rho_2) = 2/3 = c_{12}$ is exact. Physically, ρ_1 and ρ_2 correspond to the up-type and down-type sectors; the Cabibbo suppression is thus a dimension ratio on the McKay graph. Classification: **derived**.

Derivation of c_{23} : The E_8 leg dimensions are $\text{Leg}_C = N_{\text{eig}} = 9$, $\text{Leg}_B = b_1 = 6$, and $\text{Leg}_A = 11$. Then $c_{23} = \text{Leg}_A - \text{Leg}_B = a_1^2 + 1 - 2b_1 - N_{\text{eig}}$. This simplifies to $a_1^2 - 3a_1 - 10 = (a_1 - 5)(a_1 + 2)$, which vanishes identically for $a_1 = 5$. The value $c_{23} = 5$ is degenerate: $a_1 = \dim(\rho_4) = \beta_2 = \text{Leg}_A - \text{Leg}_B$. The physical expression is unambiguous (all give 5) but the selection principle is not unique. Classification: **partially derived**.

Sign rule (arm topology). The correction signs follow from arm topology on $\hat{E}_8$: paths *within* the same arm produce *negative* (screening) corrections, while paths that *cross* the branch point produce *positive* (anti-screening) corrections. Specifically, the Cabibbo mixing ($u \leftrightarrow s$) traverses the long arm only, giving the negative sign in c_{12} ; the cb mixing crosses from the medium arm (ρ_6 – ρ_7) to the short arm (ρ_8), giving a positive sign in c_{23} ; and the ub mixing crosses the widest span, with a positive sign in c_{13} . Classification: **derived**.

McKay graph adjacency. The correct adjacency of the $\hat{E}_8$ McKay graph has the branch at ρ_5 ($\dim = 6$): the medium arm runs ρ_5 – ρ_6 (τ)– ρ_7 (t), and the short arm is ρ_5 – ρ_8 (b). In particular, $\rho_7(t)$ is adjacent to $\rho_6(\tau)$ and $\rho_8(b)$ is adjacent to $\rho_5(c)$, consistent with the McKay property $2 \dim(\rho_k) = \sum_{\text{neighbors}} \dim(\rho_j)$ for all nine nodes.

Coupling selection. The coupling switch from α_s/π (for θ_{12}, θ_{23}) to $\sin^2 \theta_W$ (for θ_{13}) correlates with McKay graph distance: short-range mixing couples to QCD, long-range to electroweak. Classification: **pattern**.

S9.1 CKM Concordance Theorem: Proof

This provides the full proof of Theorem ?? (Section ?? of the main text).

Identity system. The bare CKM exponents $(n_{12}, n_{23}, n_{13}) = (3, 7, 12)$ satisfy:

$$I_1: n_{12} + n_{23} = 3 + 7 = 10 = 2a_1 \quad (\text{S37})$$

$$I_2: n_{12} \cdot n_{13} = 3 \times 12 = 36 = b_1^2 = (a_1 + 1)^2 \quad (\text{S38})$$

$$I_3: n_{23} + n_{13} = 7 + 12 = 19 = a_1^2 - a_1 - 1 \quad (\text{S39})$$

Factorisation proof. From I_1 : $n_{23} = 2a_1 - n_{12}$. From I_3 : $n_{13} = a_1^2 - a_1 - 1 - n_{23} = a_1^2 - 3a_1 + n_{12} - 1$. Substituting into I_2 :

$$n_{12}(a_1^2 - 3a_1 + n_{12} - 1) = (a_1 + 1)^2 \quad (\text{S40})$$

With $n_{12} = N_{\text{gen}} = 3$ (the Fibonacci generation gap, independently derived):

$$3(a_1^2 - 3a_1 + 2) = (a_1 + 1)^2 \implies 2a_1^2 - 11a_1 + 5 = 0 \implies (2a_1 - 1)(a_1 - 5) = 0$$

Since a_1 is a positive integer, $a_1 = 5$ uniquely. Alternatively, I_2 alone gives $(a_1 - 2)(a_1 - 5) = 0$ and I_3 gives $(a_1 - 1)(a_1 - 5) = 0$; the intersection is $a_1 = 5$.

T-matrix phase dictionary. The $\text{SU}(2)$ Chern–Simons theory at level $k = a_1 - 2 = 3$ has four objects $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}\}$ with conformal weights $h_j = j(j+1)/(k+2) = j(j+1)/a_1$. The T-matrix phases $\theta_j = e^{2\pi i(h_j - c/24)}$ produce separation gaps:

Gap	Value	Identification
$h_{1/2} - h_0 = 3/(4a_1)$	$n_{12} = 3$	N_{gen}
$h_{3/2} - h_1 = (2a_1 - 3)/(2a_1)$	$n_{23} = 2a_1 - 3 = 7$	degree $- a_1$
$h_{3/2} - h_{1/2} = (3a_1 - 3)/(2a_1)$	$n_{13} = 3a_1 - 3 = 12$	degree

The complete phase dictionary of the 4-object TQFT is $\{3, 5, 7, 8, 12, 15\} = \{N_{\text{gen}}, a_1, n_{23}, \text{rank}(E_8), \dim$ linking CKM structure to the full framework.

Discriminant. The CKM quadratic (with roots n_{23}, n_{13}) has discriminant $\Delta = (n_{13} - n_{23})^2 = (a_1^2 - 5a_1 + 3 + n_{12})^2$. For $n_{12} = 3$: $\Delta = (a_1^2 - 5a_1 + 6)^2 = ((a_1 - 2)(a_1 - 3))^2$. At $a_1 = 5$: $\Delta = (3 \cdot 2)^2 = 36$. Equivalently, $(n_{13} - n_{23})^2 = (12 - 7)^2 = 25 = a_1^2$, and the full discriminant including the linear coefficient gives $(a_1 \cdot N_{\text{gen}})^2 = 15^2 = 225$, a perfect square ensuring rational roots.

Cross-ratio. The structural identity $\phi^4 = V_{us}/V_{cb}$ (bare) follows from $\phi^{n_{23}-n_{12}} = \phi^4$, connecting the golden ratio to the CKM hierarchy. Numerically: $\phi^4 = 6.854$ vs. $V_{us}/V_{cb} = 5.47$ (bare) or 5.49 (corrected), a ratio reflecting the perturbative corrections.

Modular closure. The quotient $\text{PSL}(2, \mathbb{Z})/\Gamma(5) \cong A_5$ is the icosahedral group—the same discrete symmetry governing the 600-cell, fermion generations, and gauge structure. The CKM concordance theorem thus closes the circle: the mixing angles encode the same A_5 structure that determines all other parameters.

Honest status. The concordance is a **theorem**: the three identities are algebraically verified and their intersection uniquely selects $a_1 = 5$ (thirteenth characterisation). However, the CKM ansatz $\theta_{ij} = \arctan(\phi^{-n_{ij}} \cdot c_{ij})$ remains a **pattern**—it is not derived from the spectral action or any Lagrangian principle. The exponents are derived; the functional form is not.

S9.2 Cayley–Concordance Vacuum Offset: Proof

This provides the full proof of Theorem ?? (Section ?? of the main text).

Cayley spectral exponents. The Cayley graph Laplacian on $2I$ has eigenvalues determined by the $2I$ character table. For the two non-trivial Galois pairs (irreps exchanged by $\sigma: \sqrt{5} \rightarrow -\sqrt{5}$), the spectral ratios are:

$$\text{dim-2 pair } (\rho_1, \rho_7): \quad \lambda_t/\lambda_u = \phi^{n_{\text{ut}}}, \quad n_{\text{ut}} = \text{degree}/2 - 2 = 4 \quad (\text{S41})$$

$$\text{dim-3 pair } (\rho_2, \rho_8): \quad \lambda_b/\lambda_d = \phi^{n_{\text{db}}}, \quad n_{\text{db}} = \text{degree}/3 - 2 = 2 \quad (\text{S42})$$

The general formula $n = \text{degree}/d - 2$ for a Galois pair of dimension d is exact.

Uniform offset parameterisation. Define the Cayley→CKM map with a single offset parameter δ :

$$n_{12} = n_{\text{ut}} - \delta, \quad n_{23} = n_{\text{ut}}n_{\text{db}} - \delta, \quad n_{13} = n_{\text{ut}}(n_{\text{db}} + \delta).$$

I_3 **forces** $\delta = 1$. Substituting $n_{\text{ut}} = a_1 - 1$ and $n_{\text{db}} = (a_1 - 1)/2$ into I_3 : $n_{23} + n_{13} = a_1^2 - a_1 - 1$:

$$(a_1 - 1)^2 + \delta(a_1 - 2) = a_1^2 - a_1 - 1 \implies \delta(a_1 - 2) = a_1 - 2 \implies \delta = 1 \quad (a_1 \neq 2).$$

I_1 **gives** $a_1 = 5$. With $\delta = 1$: $n_{12} + n_{23} = (a_1 - 2) + (a_1 - 1)^2/2 - 1 = (a_1^2 - 5)/2$. Setting this equal to $2a_1$: $(a_1 - 5)(a_1 + 1) = 0$, hence $a_1 = 5$.

I_2 **confirms** $a_1 = 5$. With $\delta = 1$: $n_{12} \cdot n_{13} = (a_1 - 2)(a_1^2 - 1)/2 = (a_1 + 1)^2$ reduces to $a_1(a_1 - 5) = 0$, consistent with I_1 .

I_3 **is automatic.** With the Cayley parameterisation, I_3 is satisfied identically for all a_1 ; it encodes the *structure* of the map rather than constraining a_1 . The constraints arise from I_1 and I_2 alone—the complementary pair to the T-matrix route (where I_1 is automatic and I_2, I_3 constrain).

Non-uniform offset. If one allows three independent offsets $(\delta_1, \delta_2, \delta_3)$, the identity I_3 forces $\delta_2 = 1 + (a_1 - 1)(\delta_3 - 1)$, so $\delta_2 = 1$ iff $\delta_3 = 1$. Then I_1 gives $\delta_1 = (a_1^2 - 4a_1 - 3)/2$, which equals 1 iff $(a_1 - 5)(a_1 + 1) = 0$. Thus the *uniform offset hypothesis* $\delta_1 = \delta_2 = \delta_3$ is equivalent to $a_1 = 5$ (fourteenth characterisation).

Physical interpretation. The offset $\delta = 1$ is the standard vacuum subtraction: Cayley eigenvalues encode full spectral propagation (including the identity channel $j = 0$), while CKM angles encode only non-trivial flavor transitions ($S = 1 + iT$). In the TQFT at level $k = a_1 - 2 = 3$: the number of simple objects is $k + 1 = n_{\text{ut}} = 4$; removing the vacuum gives $k = 3 = n_{12}$.

Self-consistency: degree = 12. The constraint $n_{\text{db}} = n_{\text{ut}}/2$ implies $\text{degree}/3 - 2 = (\text{degree}/2 - 2)/2$, which gives $\text{degree} \cdot (1/3 - 1/4) = 1$, hence $\text{degree} = 12$ —the 600-cell vertex degree, uniquely determined.

Comparison of three routes.

Route	Automatic	Constraining	Factorisation
T-matrix	I_1	$I_2: (a_1 - 2)(a_1 - 5) = 0$ $I_3: (a_1 - 1)(a_1 - 5) = 0$	$\cap \Rightarrow a_1 = 5$
Cayley	I_3 (+fixes $\delta = 1$)	$I_1: (a_1 - 5)(a_1 + 1) = 0$ $I_2: a_1(a_1 - 5) = 0$	$\cap \Rightarrow a_1 = 5$
Cayley (n_{ut})	I_3	$I_1: (n_{\text{ut}} - 4)(n_{\text{ut}} + 2) = 0$ $I_2: (n_{\text{ut}} - 4)(n_{\text{ut}} + 1) = 0$	$\cap \Rightarrow n_{\text{ut}} = 4$

All three routes converge on $a_1 = 5$ via different factorisations.

S9.3 Cayley Spectral Structure and CKM Grading

This section extends the Cayley graph analysis of the previous subsections, establishing exact product rules for Galois-paired eigenvalues, the spectral origin of ϕ as the CKM base, and a $\mathbb{Z}/2$ grading that constrains the mediator structure of CKM transitions.

Exact product and sum rules. The Cayley graph of $2I$ (120 vertices, degree 12, generators from conjugacy class $10a$) has eigenvalues $\lambda_k = 12(1 - \chi_k(10a)/d_k)$ with multiplicity d_k^2 . For the two Galois pairs:

$$\lambda_u \cdot \lambda_t = (12 - 6\phi)(6 + 6\phi) = 36(2 - \phi)(1 + \phi) = 36\phi^2/\phi^2 = 36 = b_1^2, \quad (\text{S43})$$

$$\lambda_d \cdot \lambda_b = (12 - 4\phi)(8 + 4\phi) = 4(3 - \phi)(8 + 4\phi) = 80 = (a_1 - 1)^2 a_1, \quad (\text{S44})$$

where the simplification uses $\phi^2 = \phi + 1$ and $(2 - \phi)(1 + \phi) = 2 + \phi - \phi^2 = 1$. The geometric means are:

$$\sqrt{\lambda_u \lambda_t} = b_1 = 6, \quad \sqrt{\lambda_d \lambda_b} = 4\sqrt{a_1}. \quad (\text{S45})$$

Sum rules: $\lambda_u + \lambda_t = 18 = 3b_1$ and $\lambda_d + \lambda_b = 20 = 4a_1$. The differences $\lambda_t - \lambda_u = 6\sqrt{5}$ and $\lambda_b - \lambda_d = 4\sqrt{5}$ have ratio $3/2 = N_{\text{gen}}/n_{\text{db}}$.

Spectral gap and ϕ as CKM base. The spectral gap is $\lambda_{\min} = \lambda_u = 12 - 6\phi = 6/\phi^2$. Define the natural CKM time scale t_0 by $e^{-t_0 \lambda_{\min}} = 1/\phi$:

$$t_0 = \frac{\ln \phi}{\lambda_{\min}} = \frac{\phi^2 \ln \phi}{6}. \quad (\text{S46})$$

At this time scale, the heat kernel between representations decays as $e^{-t_0 \lambda_k} = \phi^{-\lambda_k/\lambda_{\min}}$. The CKM transition amplitude at “temperature” t_0 therefore involves ϕ^{-n} with $n =$

$\lambda_k/\lambda_{\min}$, explaining why ϕ appears as the natural CKM base: *it is the fundamental decay mode of the Cayley graph of $2I$.*

$\mathbb{Z}/2$ fusion grading. The fusion ring of $2I$ inherits a $\mathbb{Z}/2$ grading from the spin parity of $SU(2)$ representations:

- *Even* (integer spin): $\rho_0, \rho_2, \rho_4, \rho_6, \rho_8$ — includes down-type quarks (d, s, b) .
- *Odd* (half-integer spin): $\rho_1, \rho_3, \rho_5, \rho_7$ — includes up-type quarks (u, c, t) .

The fusion rule $\rho_i \otimes \rho_j$ preserves parity: $\text{even} \otimes \text{even} = \text{even}$, $\text{odd} \otimes \text{odd} = \text{even}$, $\text{even} \otimes \text{odd} = \text{odd}$. This is verified numerically: all $9^3 = 729$ fusion coefficients N_{ij}^k satisfy the grading with zero violations.

Proposition S4 (Single-Mediator No-Go). *For any single mediator representation ρ_W of $2I$, the CKM fusion amplitude*

$$V^{\text{fusion}}(\rho_{\text{up}}, \rho_{\text{down}}) = \sum_k N_{\text{up}, W}^k N_{k, W^*}^{\text{down}}$$

vanishes identically: $V^{\text{fusion}} = 0$ for all up-down pairs.

Proof. For ρ_W even: $\rho_{\text{up}}(\text{odd}) \otimes \rho_W(\text{even})$ produces only odd representations; $\rho_{\text{down}}(\text{even}) \otimes \rho_{W^*}(\text{even})$ produces only even representations. The sets of intermediate representations have disjoint parity, so the sum vanishes. For ρ_W odd: $\rho_{\text{up}} \otimes \rho_W$ produces even intermediates; $\rho_{\text{down}} \otimes \rho_{W^*}$ produces odd intermediates. Again disjoint. $\square$

Consequence. CKM mixing requires at minimum a *two-mediator* process, consistent with the Standard Model where the W boson couples through the fundamental representation ρ_1 (odd, dimension 2) and CKM mixing arises from the Yukawa sector, not from gauge boson exchange alone. The $\mathbb{Z}/2$ grading is the discrete residue of the continuous $SU(2)$ isospin symmetry within the finite group $2I$.

Assessment. The Cayley graph of $2I$ explains *why ϕ appears* in CKM mixing (spectral gap) and provides exact eigenvalue product rules connecting b_1 and a_1 to the Galois-paired quarks. However, the specific CKM exponents $\{3, 7, 12\}$ do not emerge from the Cayley graph dynamics directly; they require either the algebraic concordance identities (Theorem ??), the vacuum offset (Theorem ??), or the E_8 decomposition overlap structure (Corollary ??; see Section S9.4 below). The functional form $\theta_{ij} = \arctan(\phi^{-n_{ij}} \cdot c_{ij})$ remains an ANSATZ: derivation from a Lagrangian principle is an open problem.

S9.4 E_8 Decomposition Overlap and CKM: Third Route

This expands on the proof of Theorem ?? and Corollary ?? in the main text.

S9.4.1 The 40 decompositions

The E_8 root system (240 roots in $\mathbb{R}^8$) projects onto 4-dimensional subspaces via Coxeter element eigenspace decomposition. Each Coxeter element $c \in W(E_8)$ has order $h = a_1 b_1 = 30$ and eigenvalues $e^{2\pi i m/h}$ for the Coxeter exponents $m \in \{1, 7, 11, 13, 17, 19, 23, 29\}$. The H_4 subspace is the real span of the eigenplanes for exponents $\{1, 11, 19, 29\}$ (the quadratic residues mod a_1); the complementary H'_4 uses $\{7, 13, 17, 23\}$ (the non-residues).

Projecting the 240 roots onto such an H_4 subspace splits them into two sets of 120: the *inner* 600-cell (projected $\text{norm}^2 = 1 - 1/\sqrt{a_1}$) and the *outer* 600-cell (projected $\text{norm}^2 = 1 + 1/\sqrt{a_1}$), with norm ratio ϕ . Exhaustive enumeration over all $8!$ orderings of simple reflections yields exactly $40 = \text{rank}(E_8) \cdot a_1$ distinct decompositions.

S9.4.2 Proof of the quantization

The H_4 projector admits a character expansion:

$$P = \frac{2}{h} \sum_{k=0}^{h-1} F(k) c^k, \quad F(k) = \cos\left(\frac{2\pi k}{h}\right) + \cos\left(\frac{2\pi \cdot 11 k}{h}\right). \quad (\text{S47})$$

Using the product-to-sum identity with $h = a_1 b_1$:

$$F(k) = 2 \cos\left(\frac{2\pi k}{a_1}\right) \cos\left(\frac{\pi k}{3}\right) =: 2 A(k) B(k), \quad (\text{S48})$$

where $A(k) = \cos(2\pi k/a_1)$ has period a_1 and $B(k) = \cos(\pi k/3)$ has period $b_1 = 6$.

The Grassmannian trace is:

$$\text{Tr}(P_1 P_2) = \frac{4}{h^2} \sum_{j,k=0}^{h-1} F(j) F(k) T(j, k), \quad T(j, k) := \text{Tr}(c_1^j c_2^k) \in \mathbb{Z}. \quad (\text{S49})$$

The Chinese Remainder Theorem ($h = a_1 b_1$, $\gcd(a_1, b_1) = 1$) bijects $j \leftrightarrow (a, u)$ with $a = j \bmod a_1$, $u = j \bmod b_1$. Define:

$$\sigma(a, b) := \sum_{u,v \in \mathbb{Z}/b_1} B(u) B(v) T(j(a, u), k(b, v)). \quad (\text{S50})$$

Step 1 (Rationality of σ). Since $B(u) \in \{1, \frac{1}{2}, -\frac{1}{2}, -1\}$ and $T \in \mathbb{Z}$, we have $4\sigma(a, b) \in \mathbb{Z}$.

Step 2 (Galois symmetry). The B -weighted sum filters the trace to its fundamental Fourier mode on $\mathbb{Z}/b_1$. The resulting $\sigma(a, b)$ is the real part of a complex expression involving $e^{2\pi i m a/a_1}$. The Galois involution $a \rightarrow -a \pmod{a_1}$ acts by complex conjugation, and since $\text{Re}(z) = \text{Re}(\bar{z})$:

$$\sigma(a, b) = \sigma(-a \bmod a_1, -b \bmod a_1). \quad (\text{S51})$$

This is verified numerically: individual $T(j, k)$ values are *not* Galois-invariant (~ 500 mismatches per pair), but $\sigma(a, b)$ is (to machine precision, $\sim 10^{-14}$, for all $\binom{40}{2} = 780$ pairs).

Step 3 (Irrational cancellation). The $A(a)$ values involve $\sqrt{5}$ for $a \neq 0$: $A(a) = \cos(2\pi a/5)$ takes values $1, (\sqrt{5}-1)/4, -(\sqrt{5}+1)/4$. The Galois symmetry of σ forces the $\sqrt{5}$ coefficient in $\sum_{a,b} A(a) A(b) \sigma(a, b)$ to vanish identically, leaving $\text{Tr}(P_1 P_2) \in \mathbb{Q}$.

Step 4 (Denominator). With $\sigma \in \mathbb{Z}/4$, A -products contributing denominator at most 8, and the overall factor $16/h^2 = 16/900$, the denominator of $\text{Tr}(P_1 P_2)$ divides a_1 . Combined with overlap $= h \cdot \text{Tr} \in \mathbb{Z}$, the overlap is a multiple of $h/a_1 \cdot 2 = 2b_1 = 12$.

S9.4.3 Range restriction

The overlap levels $m \in \{5, 6, 7, 8\}$ are constrained by:

Upper bound. For the closest pair of distinct H_4 subspaces, three of four Coxeter eigenplanes align perfectly ($\cos^2 = 1$) and the fourth has the minimum non-zero angle $\cos^2 = 1/a_1$. This gives $m_{\max} = a_1(3 + 1/a_1)/2 = (3a_1 + 1)/2$. The identity $(3a_1 + 1)/2 = \text{rank}(E_8) = 8$ is equivalent to $a_1 = 5$.

Lower bound. For pairs sharing two or more eigenplanes (652 of 780), $\sum \cos^2 \geq 2 = 2a_1/a_1$, forcing $m \geq a_1 = 5$. The remaining 128 pairs (with one approximate shared direction) satisfy $m = 5$ by explicit computation.

The self-duality $\text{Tr}(P_1 P_2) = \text{Tr}((I - P_1)(I - P_2))$ (which follows from $\text{rank}(P) = \text{rank}(I - P) = 4$) ensures that the H_4 and H'_4 overlap structures are identical.

S9.4.4 CKM derivation from overlaps

Given the four overlaps $\{60, 72, 84, 96\}$:

1. $n_{13} = \gcd(60, 72, 84, 96) = 12$. This is the *fundamental exchange quantum*: roots switch between 600-cells in multiples of 12 (the vertex degree).
2. The m -values $\{5, 6, 7, 8\}$ and multipliers $\{5, 4, 3, 2\} = \{a_1, n_{\text{ut}}, n_{12}, n_{\text{db}}\}$ are related by the involution $\sigma: m \mapsto 2a_1 - m$, which pairs $n_{23} = 7 \leftrightarrow n_{12} = 3$. The concordance identity I_1 ($n_{12} + n_{23} = 2a_1$) is this duality.
3. Among the candidates $n_{23} \in \{6, 7, 8\}$, only $n_{23} = 7$ satisfies the concordance identity $I_2: n_{12} \cdot n_{13} = 3 \cdot 12 = 36 = b_1^2$.

The sum of m -values is $5 + 6 + 7 + 8 = 26 = a_1^2 + 1 = 1/\sin^2 \theta_W$, connecting the overlap spectrum to the Weinberg angle.

Reproducibility. The verification script `verify_e8_decomposition_ckm.py` (in the supplementary code repository) independently constructs the E_8 root system, finds all 40 decompositions, computes all 780 pairwise overlaps, and checks 34 assertions covering quantization, CKM multipliers, Galois cancellation, root counting, and algebraic identities. All 34 tests pass.

S10 CP: Majorana Phases

This expands on Section 9 of the main text.

For Majorana neutrinos, the PMNS matrix carries two additional phases α_1, α_2 (unmeasured):

$$U_{\text{PMNS}} = U_{\text{Dirac}} \cdot \text{diag}(1, e^{i\alpha_1/2}, e^{i\alpha_2/2}) \quad (\text{S52})$$

The three neutrino mass eigenstates in ρ_3 of $2I$ correspond to angular momentum projections $m = -1, 0, +1$, with eigenvalues ζ_5^m under a 5-fold element. The Frobenius automorphism $\sigma_2: \zeta_5 \rightarrow \zeta_5^2$ shifts the eigenvalue phase:

$$\alpha_1 = \frac{4\pi}{a_1} = 144^\circ, \quad \alpha_2 = -\frac{4\pi}{a_1} = -144^\circ \quad (\text{S53})$$

Testability. The effective Majorana mass for neutrinoless double beta decay is $|m_{ee}| \approx 2.6$ meV.

Caveat. The assignment and the identification of the Frobenius twist with the Majorana phase are motivated but not uniquely derived. **Category: partially derived.**

S11 The Galois Dark Sector

This section presents the full dark sector analysis. A summary appears in Section 13 of the main text.

S11.1 Galois Partner Masses

For each fermion with $m_f = m_e \cdot \phi^n$, the Galois partner has mass

$$|m_{f'}| = m_e \cdot |\phi'^n| = \frac{m_e}{\phi^n} \implies m_f \cdot |m_{f'}| = m_e^2 \quad (\text{mass duality}) \quad (\text{S54})$$

Fermion	(a, b)	n	m_f (MeV)	$ m_{f'} $ (keV)	Scale
e'	(0, 0)	0	0.511	511.0	sub-MeV
u'	(3, -2)	3	2.16	120.7	sub-MeV
d'	(1, 0)	5	5.67	46.1	keV
μ', s'	(1, 1)	11	102/94	2.57	keV
c'	(2, 1)	16	1 128	0.232	sub-keV
τ'	(1, 2)	17	1 825	0.143	sub-keV
b'	(-1, 4)	19	4 777	0.0547	sub-keV
t'	(4, 1)	26	138 700	0.00188	eV

S11.2 Galois Coupling Constants

Weinberg angle. $\sin^2 \theta_W = 6/26$ is rational, hence Galois-invariant: *identical* in both sectors.

Strong coupling. $\alpha'_s = \phi' - 3/2 < 0$. A *negative* QCD coupling means **no confinement**: Galois quarks are free.

Fine structure constant. The Galois equation has negative discriminant: α' is complex, and the Galois sector has **no real electromagnetic coupling**.

S11.3 Dark Matter Properties

The Galois sector has exactly the properties required of dark matter:

1. **Electromagnetically dark:** α' is complex (derived).
2. **Weakly dark:** complex α' (derived).
3. **QCD-deconfined:** $\alpha'_s < 0$ (derived).
4. **Gravitationally interacting only** (derived).
5. **Absolutely stable:** no real gauge couplings (derived).
6. **Multi-component:** 9 stable species with 8 distinct masses.

S11.4 Galois Neutrinos: the Inverted Seesaw

Under Galois conjugation, the seesaw is *inverted*: tiny Majorana mass yields *super-heavy* neutrinos $m_{\nu'} \sim 10^{11}\text{--}10^{14}$ GeV.

S11.5 The Dark Matter Abundance

The observed ratio $\Omega_{\text{DM}}/\Omega_b = 5.36 \pm 0.05$ is matched by

$$\frac{\Omega_{\text{DM}}}{\Omega_b} = 7 - \phi = 5.382 \quad (\text{error } 0.41\%) \quad (\text{S55})$$

Caveat: this is a **pattern**, not a derivation.

Gravitational freeze-in. The only viable production mechanism is gravitational freeze-in. The mass budget is dominated by e' (75%), u' (18%), d' (7%).

Remark (reheating scale). The required T_{reh} is well approximated by $T_{\text{reh}} = M_{\text{Pl}}/\phi^{a_1+b_1}$. This is classified as a **pattern**.

S12 Topological Entanglement and Quantum Groups

This section presents the full TQFT analysis. The bootstrap theorem (Theorem 1, Section 1 of the main text) establishes the TQFT foundation; a summary of entanglement results appears in Section 13.

S12.1 Fibonacci Anyons from a_1

$SU(2)$ Chern–Simons theory at level $k = a_1 - 2 = 3$ admits representations $j = 0, \frac{1}{2}, 1, \frac{3}{2}$ with quantum dimensions $\{1, \phi, \phi, 1\}$, where the Fibonacci anyon τ has

$$d_\tau = 2 \cos\left(\frac{\pi}{a_1}\right) = \phi \quad (\text{S56})$$

The fusion rule $\tau \times \tau = \mathbf{1} + \tau$ is the defining relation $\phi^2 = 1 + \phi$ of the golden ratio. The *Verlinde fusion ring* of $SU(2)_3$ is isomorphic to $\mathbb{Z}[\phi]/(\phi^2 - \phi - 1)$.

S12.2 Galois Conjugation as Quantum Group Duality

The Galois automorphism σ acts as $q \rightarrow q^2$ on the Chern–Simons side. The Galois conjugate theory has quantum dimensions $\{1, 1/\phi, -1/\phi, -1\}$ —non-unitary, paralleling $\alpha'_s < 0$ and complex α' .

S12.3 Total Quantum Dimension and Entropy

$$D^2 = a_1 + \sqrt{a_1} = 5 + \sqrt{5}, \quad D'^2 = a_1 - \sqrt{a_1} = 5 - \sqrt{5}, \quad (\text{S57})$$

$$D^2 + D'^2 = 2a_1 = 10, \quad D^2 \cdot D'^2 = a_1(a_1 - 1) = 20 = \frac{N}{b_1} \quad (\text{S58})$$

The topological entanglement entropy difference between the physical and dark sectors is exactly $\ln \phi$.

S12.4 McKay Graph Entanglement

For the tight-binding Hamiltonian $H = -A$ at half-filling, the entanglement entropy of the chirality bipartition is $S_{\text{McKay}} = 4 \ln 2 = \ln[(a_1 - 1)^2]$.

S13 The Cosmological Constant

This section presents the full CC analysis. A summary appears in Section 13 of the main text.

S13.1 The Exponent Formula

$$z_\Lambda = \frac{N - N_{\text{gen}}}{2} - \phi = 56.882 \quad (\text{S59})$$

equivalently $\Lambda_P = \alpha^{57-1/(2\phi^3)}$. Matches observation to 0.13σ .

S13.2 Seven Decompositions of 57

$$57 = N/2 - N_{\text{gen}} = 60 - 3 \quad (\text{geometric}) \quad (\text{S60})$$

$$= \sum_{k=1}^9 z_k + (b_1 + 1) = 50 + 7 \quad (\text{spectral}) \quad (\text{S61})$$

$$= N_{\text{gen}} \cdot N(a_1 + b_1\phi) = 3 \times 19 \quad (\text{algebraic}) \quad (\text{S62})$$

$$= \frac{1}{2} \sum_{f=1}^9 n_f + N_{\text{gen}} = 54 + 3 \quad (\text{mass spectrum}) \quad (\text{S63})$$

$$= |\eta_A| + \sum_{i < j} n_{ij}^{\text{CKM}} = 35 + 22 \quad (\text{bridge}) \quad (\text{S64})$$

$$= \sum_{\lambda_k < 0} d_k^2 - \text{rank}(E_8) = 65 - 8 \quad (\text{mode counting}) \quad (\text{S65})$$

$$= \Lambda_{\text{UV}}^2 R^2 + (h(E_8) - \text{rank}(E_8)) = 35 + 22 \quad (\text{UV cutoff}) \quad (\text{S66})$$

S13.3 Galois Duality

$\Lambda_P \cdot \Lambda'_P = (\alpha \alpha')^{57-\alpha_s} = (2\pi)^{-(57-\alpha_s)}$. The visible CC is tiny *because* the Galois conjugate is huge.

S13.4 The Neutrino–CC Connection

$\Lambda^{1/4} \approx m_3/(8\phi^2 + 1) = 2.26 \text{ meV}$ (obs: 2.24 meV, error 0.79%).

S13.5 Spectral Determinant Test

The Cayley graph Laplacian $L_k = 12(1 - \chi_k(C_{10})/d_k)$ has nine eigenvalues in $\mathbb{Z}[\phi]$ with multiplicities d_k^2 , and zero mode $L_0 = 0$ (multiplicity 1). Its regularized determinant

$$\det'(L) = \prod_{k: L_k > 0} L_k^{d_k^2} \quad (\text{S67})$$

is a finite, well-defined product (no UV divergences on a finite graph). One might hope that $\log \det'(L)/\ln \alpha = z_\Lambda$, thereby deriving the functional form α^z .

Result: With the normalization $L \rightarrow L/N$ (the only one that brings the ratio near the target),

$$\frac{\log \det'(L/N)}{\ln \alpha} = 56.97, \quad (\text{S68})$$

which differs from 57 by 0.026 and from $z_\Lambda = 56.88$ by 0.092. No principled normalization closes the gap: the required divisor $c = 119.54$ (vs. $N = 120$) has no geometric justification. The heat kernel $K(t) = \sum_k d_k^2 e^{-tL_k} \rightarrow 1$ as $t \rightarrow \infty$ (zero-mode floor), so $\alpha^{57} \sim 10^{-122}$ is unreachable. The Reidemeister torsion $\log T_R = 0$ exactly (as required for S^3 with trivial representation). The spectral zeta function $\zeta_L(s) = \sum_{k>0} d_k^2 L_k^{-s}$ achieves no integer or framework value at any special s .

Conclusion: The functional form α^z is *not* derivable from the Cayley graph spectral determinant.

Theorem S3 (Galois invariance of the spectral determinant). $\det'(L) \in \mathbb{Q}$. *Equivalently, $\sigma(\det'(L)) = \det'(L)$ under the Galois automorphism $\sigma : \phi \mapsto \phi' = -1/\phi$.*

Proof. The Galois action permutes the irrational eigenvalues in pairs: $\sigma(L_1) = L_7$ and $\sigma(L_2) = L_8$, where $L_1 = 12-6\phi$, $L_7 = 6+6\phi$, $L_2 = 12-4\phi$, $L_8 = 8+4\phi$. The corresponding irreps ρ_2 and ρ_8 both have dimension 2 (hence multiplicity 4), while ρ_3 and ρ_9 both have dimension 3 (hence multiplicity 9). The rational eigenvalues (L_3, L_4, L_5, L_6) are Galois-fixed. Therefore $\sigma(\det'(L)) = \prod_{k>0} \sigma(L_k)^{d_k^2} = \prod_{k>0} L_{\sigma(k)}^{d_k^2} = \prod_{k>0} L_k^{d_k^2} = \det'(L)$, where the penultimate equality uses $d_k = d_{\sigma(k)}$ to reindex the product. $\square$

The Galois norms $N(L_k) = L_k \cdot \sigma(L_k)$ are 36, 80, 81, 144, 196, 225 (for $k = 1, \dots, 6$ with $N(L_7) = N(L_1) = 36$, $N(L_8) = N(L_2) = 80$), all perfect squares or small composites. The rationality of $\det'(L)$ is a structural consequence of the 2I representation theory—Galois-conjugate irreps have equal dimensions—not a numerical coincidence.

Category: partially derived. The exponent 57 is fully derived; the base α^z remains a pattern.

S14 Cosmological Consistency and Inflation

This section presents the full cosmological analysis. A summary appears in Section 13 of the main text.

S14.1 R^2 Inflation from the Spectral Action

The Chamseddine–Connes spectral action on the product geometry generates Starobinsky R^2 inflation with $N_e = N/2 = 60$. Predictions: $n_s = 0.9654$, $r = 0.0033$, both within Planck 2018 bounds.

S14.2 Vacuum Stability and Leptogenesis

The framework top mass $m_t = 172.5$ GeV places the electroweak vacuum near the stability boundary. Leptogenesis is viable since $T_{\text{reh}} \gg M_R$.

S14.3 The Scalon Mass and the Test Function Problem

The scalaron mass depends on f_4 , which is genuinely free in any NCG model. A numerical pattern: $f_4 = N^4/(2\pi)$ gives $A_s = 2.116 \times 10^{-9}$ (+0.8%, 0.05 σ). **Category: pattern.**

S15 Complete Master Table with Footnotes

See Table S3 for the complete prediction summary with detailed footnotes.

Quantity	Error	Status	Free params
<i>Coupling constants</i>			
α (all coefs from a_1 ; ϕ^4 structural)	0.0001%	Structural ^a	0
α_s	0.11%	Derived	0
$\sin^2 \theta_W$	0.19%	Derived	0
$a_\mu(\text{NP}) = 0$	—	Derived	0
<i>Boson masses and Planck scale</i>			
m_Z	0.09%	Derived (+running)	0 (+ m_e)
m_H/m_W	0.20%	Derived	0 (+ m_W^{exp})
m_e/m_P	0.24%	Derived	0 (+ m_e)
<i>Gravity</i>			
$\gamma_{\text{disc}} = 1$ (discrete scalar response)	Exact	Derived [‡]	0
Gravity eq. $Ch_T = 8\pi G T_T$	—	Derived	0
Gap-Planck: $\lambda_1^{\text{ex}} \cdot 4\phi^2 = 2T $	Exact	Derived	0
Hierarchy: $z_{\text{Planck}} = 1/\alpha_s + 2$	Exact	Derived	0
Spectral action $S = \text{Tr } f(D/\Lambda)$	—	Derived	0
<i>Structural</i>			
$N_{\text{gen}} = 3$	Exact	Derived	0
Gauge: $SU(3) \times SU(2) \times U(1)$	Exact	Derived	0
(a, b) quantum numbers unique	Exact	Derived	0
McKay sum rules $\sum a = 12, \sum b = 8$	Exact	Derived	0
$ G(\rho_4) = N_{\text{gen}}$ (spectral)	Exact	Derived	0
<i>Fermion masses (with one-loop QED)</i>			
m_μ, m_τ, m_u (bare)	0.2–3.8%	Derived	0 (+ m_e)
m_c, m_t, m_d, m_s, m_b (bare)	9–21%	Derived	0 (+ m_e)
μ (Galois + one-loop)	+2.57 σ	Derived	0 (+ m_e)
τ (Galois + one-loop)	−0.96 σ	Derived	0 (+ m_e)
b (norm-log + one-loop)	−1.49 σ	Derived	0 (+ m_e)
All 9 fermions combined	$\chi^2/\text{dof} = 1.26$	Derived	0 (+ m_e)
<i>Mixing angles</i>			
CKM ($\theta_{12}, \theta_{23}, \theta_{13}$)	< 0.2%	Derived	0
PMNS (θ_{23}, θ_{13})	< 0.2%	Derived	0
PMNS (θ_{12})	2.3% (0.8 σ)	Derived	0
<i>CP violation</i>			
$\delta_{\text{CKM}} = \arctan(\sqrt{5})$	0.77%	Derived	0
$\delta_{\text{PMNS}} = 3 \arctan(\sqrt{5})$	0.36%	Derived	0
$\theta_{\text{QCD}} = 0$	Exact	Derived	0
$J = 3.12 \times 10^{-5}$	1.1%	Derived	0
$\alpha_{1,2} = \pm 4\pi/5$ (Majorana)	—	Partially derived	0
<i>Neutrinos</i>			
m_3	< 1%	Derived	0 (+ m_e)
$\Delta m_{21}^2/\Delta m_{31}^2 = \alpha\phi^3$	5%	Derived	0
η_B (baryogenesis)	$\sim 2\times$	Pattern	0
<i>Dark sector</i>			
Galois masses, stability, darkness	Exact	Derived	0
$\Omega_{\text{DM}}/\Omega_b = 7 - \phi$	0.41%	Pattern	0
<i>Cosmological constant</i>			
$\Lambda_P = \alpha^{57-\alpha_s}$	0.33%	Pattern	0

Table S3: Complete summary of all predictions with derivation details. “Derived” means deduced from $a_1 = 5$ via established mathematics. “Partially derived” means the geometric components are identified but a gap remains. “Pattern” means algebraic formula matching experiment but not uniquely derived.

^aBoth coefficients from icosahedral Laplacian eigenvalues: linear = product $L(\mathbf{3})L(\mathbf{3}')$, quadratic = sum $[L(\mathbf{3}) + L(\mathbf{3}')] \cdot \pi/a_1$ (decagonal holonomy = 2π , exact geometric fact; entrance into coupling equation via Kaluza–Klein identification).

^bFactor structure $z = c\phi^{N_{\text{gen}}}$ with $\text{Tr}(z) = 8$ gives $c = 2$, $|N| = 4 = N/h$ uniquely.

S16 Uniqueness and Variational Selection

This expands on Section 14 of the main text.

S16.1 Uniqueness Among Regular Polytopes

A recurrent objection is that $\mathbb{Z}[\phi]$ and E_8 are “so rich that any regular polytope would produce something resembling the Standard Model.” We refute this by systematic enumeration: there are exactly six regular polytopes in four dimensions, and we test each against seven criteria required by the framework.

Polytope	Ring	a_1	T1	T2	T3	T4	T5	T6	T7
5-cell	$\mathbb{Q}$	1	×	×	×	×	×	×	×
8-cell	$\mathbb{Z}[\sqrt{2}]$	2	×	×	×	×	×	×	×
16-cell	$\mathbb{Z}[\sqrt{2}]$	2	×	×	×	×	×	×	×
24-cell	$\mathbb{Z}[\omega]$	3	×	×	×	×	×	×	×
120-cell	$\mathbb{Z}[\phi]$	5	✓	×	✓	✓	✓	✓	✓
600-cell	$\mathbb{Z}[\phi]$	5	✓	✓	✓	✓	✓	✓	✓

Table S4: Seven tests applied to all six regular 4D polytopes: T1 = $N_{\text{gen}} = 3$, T2 = gauge group G_{SM} , T3 = $\sin^2 \theta_W \approx 0.231$, T4 = $\alpha^{-1} \approx 137$, T5 = fermion mass hierarchy, T6 = CKM/PMNS mixing, T7 = anomaly cancellation. Only the 600-cell passes all seven.

The tests are: (T1) the ring of integers provides exactly $N_{\text{gen}} = 3$ generations via $|\mathcal{N}(1 + b\phi)| = 1$; (T2) the vertex degree decomposes as $12 = 1 + 3 + 3' + 5$ under the A_5 action; (T3) $\sin^2 \theta_W = b_1/(a_1^2 + 1) \approx 0.231$; (T4) the spectral equation for α gives $\alpha^{-1} \approx 137$; (T5) the algebraic number $\phi > 1$ generates mass hierarchy via ϕ^n ; (T6) the McKay correspondence $2I \rightarrow E_8$ provides CKM/PMNS mixing; (T7) the bipartite McKay graph gives 16 Weyl fermions per generation with anomaly cancellation.

Why every alternative fails. The four genuinely independent polytopes (5-cell, 8/16-cell, 24-cell) all score 0/7. Three failure modes are decisive:

- *Foundational:* The equation $a_1! = 4a_1(a_1 + 1)$ has $a_1 = 5$ as its unique positive integer solution ($5! = 120 = 4 \cdot 5 \cdot 6$). For $a_1 = 3$ (24-cell): $3! = 6 \neq 48$; for $a_1 = 2$ (8/16-cell): $2! = 2 \neq 24$. Without this identity, no subsequent formula is justified.
- *Algebraic:* The 24-cell’s ring $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$) gives $|\omega| = 1$, so ω^n has unit modulus for all n . This makes mass hierarchy *impossible*: $m_f/m_e = |\omega|^n = 1$ for every fermion. In contrast, ϕ^n grows exponentially, spanning five orders of magnitude. Similarly, $\mathbb{Z}[\sqrt{2}]$ gives $N_{\text{gen}} = 2$ (not 3), since $|1 - 2b^2| = 1$ has only $b \in \{-1, 0, 1\}$.
- *Geometric:* The Weinberg angle formula gives $\sin^2 \theta_W = 0.40$ for the 24-cell (73% deviation) and 0.60 for the 8/16-cell (159%); $\alpha_s = 1/(2\sqrt{2}^3) = 0.177$ misses by 50% for $\mathbb{Z}[\sqrt{2}]$.

The 120-cell test. The 600-cell’s dual, the 120-cell, shares the *same* algebraic structure ($a_1 = 5$, $\mathbb{Z}[\phi]$, $2I \rightarrow E_8$) and passes 6/7 tests. It fails only T2: its vertex degree is 4 (tetrahedron), which decomposes as $4 = 1 + 3$ with no room for the color octet. This is a *strengthening* of the argument: even the polytope with identical algebra fails when

the geometry (vertex figure) is wrong. Both the algebraic ring *and* the icosahedral vertex figure are necessary; neither suffices alone.

The conclusion is unambiguous: among all six regular 4D polytopes, the 600-cell is the *unique* one that can reproduce the Standard Model. The framework is not “infinitely accommodating”—it is rigidly selective.

S16.2 Variational Selection: Minimum Action over ADE

The polytope comparison tests only six geometries. A stronger test asks: among *all* finite subgroups of $SU(2)$ —the complete ADE classification—does the binary icosahedral group $2I$ emerge from a variational principle?

Definition 1. For a finite subgroup $G \subset SU(2)$ with McKay graph Γ_G (adjacency matrix A_G , group order $|G|$), define the *spectral action density*

$$\rho(G) := \frac{\text{Tr}(A_G^2)}{|G|}. \quad (\text{S69})$$

Since $\text{Tr}(A^2) = 2|E|$ for any simple graph (with $|E|$ the number of edges), and the McKay graphs of D- and E-type are trees with $|E| = n - 1$ (where n is the number of nodes = irreducible representations), while A-type graphs are cycles with $|E| = n$, and $|G| = \sum_i d_i^2$ by character orthogonality, we obtain closed forms:

Theorem S4 (Variational selection of $2I$). *Among all finite subgroups G of $SU(2)$, the spectral action density $\rho(G)$ is uniquely minimized by $G = 2I$:*

$$\rho(2I) = \frac{2 \cdot 8}{120} = \frac{2}{15} = \frac{2}{a_1 \cdot N_{\text{gen}}} < \rho(G) \quad \text{for all } G \neq 2I.$$

Proof. The three families are handled separately.

A-type ($G = \mathbb{Z}_n$): the McKay graph is a cycle, $|E| = n$, $|G| = n$, so $\rho = 2$ for $n \geq 3$ (and $\rho = 1$ for $n = 2$).

D-type ($G = 2D_n$, $n \geq 2$): $\rho = 2(n + 2)/(4n) = \frac{1}{2} + \frac{1}{n} > \frac{1}{2}$.

E-type (binary polyhedral groups of $\{3, q\}$, $q = 3, 4, 5$): Euler’s formula on S^2 with triangular faces gives $|G_{\text{binary}}| = 24q/(6 - q)$. The McKay correspondence assigns $G \leftrightarrow \hat{E}_{q+3}$, a tree with rank = $q + 3$ edges. Therefore

$$\rho(q) = \frac{2(q + 3)}{24q/(6 - q)} = \frac{(q + 3)(6 - q)}{12q}. \quad (\text{S70})$$

The derivative $\rho'(q) = -(q^2 + 18)/(12q^2) < 0$ for all $q > 0$, so ρ is *strictly decreasing* in q . Since $q \leq 5$ by Gauss–Bonnet (angular defect $2\pi - q\pi/3 > 0$ requires $q < 6$), the minimum over E-types is $\rho(5) = 8 \cdot 1/60 = 2/15$.

Assembling: $\frac{2}{15} < \frac{7}{24} < \frac{1}{2} \leq$ all D-types $\leq$ all A-types, so the global minimum is unique. $\square$

Remark 1 (Geometric interpretation). The angular defect per vertex of $\{3, q\}$ is $\delta(q) = \pi(6 - q)/3$, which decreases with q . The icosahedron ($q = 5$, $\delta = \pi/3$) has the smallest defect among Platonic solids—it is the regular polyhedron closest to a flat tiling of S^2 . Minimizing ρ is thus equivalent to selecting the most uniformly curved finite geometry: nature chooses the discrete structure with the least concentrated curvature.

The value $\rho(2I) = 2/(a_1 N_{\text{gen}}) = 2 \cdot \text{rank}(E_8)/|2I|$ connects the variational minimum to both the framework parameter $a_1 = 5$ and the Lie-algebraic data of E_8 . Equivalently, $2I$ maximizes the *group order per edge*:

$$\frac{|G|}{|E|} = \frac{120}{8} = 15 = a_1 \cdot N_{\text{gen}},$$

more than double the runner-up ($2O$: $48/7 \approx 6.9$).

Physical interpretation. The spectral action density $\rho(G)$ is the leading term of $\text{Tr}(f(D^2))/|G|$ in the Seeley–DeWitt expansion, normalized by the discrete volume $|G|$. Minimizing ρ is the discrete analogue of the principle of stationary action: *nature selects the finite geometry that minimizes the spectral action per unit volume*. This upgrades the uniqueness from a diagnostic observation (the 600-cell *happens* to match the Standard Model) to a dynamical selection principle (the 600-cell is *forced* by action minimization).

Combined with the TQFT bootstrap (Theorem 1 of the main text), the selection is doubly constrained: $a_1 = 5$ is the unique self-consistent vacuum *and* the unique action minimum.

S16.3 Related Work

Several programs share the goal of deriving Standard Model structure from mathematical first principles; we compare the present framework with the most closely related ones.

Chamseddine–Connes–Marcolli (NCG). The spectral action programme [3, 4] derives the SM Lagrangian coupled to gravity from a product spectral triple $M^4 \times F$, where F is a finite noncommutative space. Our framework adopts the same dynamical principle (Section 12 of the main text), but the finite geometry is *determined* by $a_1 = 5$ —the McKay graph of $2I$ (affine $\hat{E}_8$)—rather than chosen to match the SM. Key differences: (i) the internal chirality $\gamma_F = (-1)^{2j}$ is derived from the bipartite structure, not selected by hand; (ii) the Yukawa couplings have zero free parameters ($\|Y\|_F = 1/\sqrt{2}$ universally); (iii) $N_{\text{gen}} = 3$ is proved, not input. Both frameworks share the limitation that the M^4 factor is not derived.

Boyle–Farnsworth. Boyle and Farnsworth showed that the finite NCG geometry of the Standard Model can be compressed to a remarkably small set of algebraic axioms, raising the question of *why* this particular algebra is selected. Our answer is that the 600-cell and the McKay correspondence with E_8 single out a unique finite geometry, providing the missing selection principle.

Furey–Hughes (division algebras). Furey and Hughes derive three SM generations from the triality symmetries of $\mathbf{tri}(\mathbb{C}) \oplus \mathbf{tri}(\mathbb{H}) \oplus \mathbf{tri}(\mathbb{O})$. The present framework arrives at $N_{\text{gen}} = 3$ from an independent route (Fibonacci–Galois on the icosahedron), but the two approaches share a structural feature: both exploit the D_4 triality symmetry—theirs via the division algebra chain, ours via the branching node of the E_8 McKay graph.

Todorov–Drenska (exceptional structures). Todorov and Drenska use the exceptional Jordan algebra $J_3(\mathbb{O})$ and its automorphism group F_4 to organise the three SM generations. Their approach and ours share the idea that the generation structure originates from an exceptional algebraic object ($J_3(\mathbb{O})$ vs. $2I/E_8$), and both exploit the 3×16

fermion counting. The 600-cell framework additionally derives coupling constants and mass ratios, which the Jordan algebra approach does not address.

Singh (Jordan algebra masses). Singh obtains fermion mass relations from $J_3(\mathbb{O}_\mathbb{C})$, including $\sqrt{m_\tau/m_\mu} = \sqrt{m_s/m_d}$. In our framework this relation is *exactly* satisfied—both ratios equal ϕ^3 , because $n_\tau - n_\mu = n_s - n_d = b_1 = 6$. This provides an independent confirmation: the common ϕ^3 factor is a consequence of the 600-cell mass formula, not a fit.

Wilson (E_8 embedding uniqueness). Wilson [7] proves that the discrete symmetries of the Standard Model (weak hypercharge, generation symmetry, colour) can be embedded in $E_8(-24)$ in a *unique* way, leading to an automatically chiral model with three generations. Two overlaps with the present framework are notable. First, Wilson’s model predicts $\delta_{\text{PMNS}} \approx 200^\circ$ (stated as a conjecture without rigorous derivation); our framework derives $\delta_{\text{PMNS}} = 3 \arctan(\sqrt{5}) = 197.7^\circ$ from the Galois inner product, within 1.2% of Wilson’s value—suggesting both approaches may access the same underlying geometric structure. Second, chirality arises independently in both frameworks: Wilson obtains it from the non-compact real form $E_8(-24)$ with quaternionic tensor products, while we derive it from the bipartite grading $\gamma_F = (-1)^{2j}$ of the McKay graph. Two independent routes to the same chirality mechanism strengthen the case that chirality is a consequence of E_8 , not an input. A key difference is that Wilson’s model does not derive coupling constants or mass ratios, while the present framework does not establish the $E_8(-24)$ embedding rigorously.

S17 Numerical Calculations

This section contains the full proofs and extended discussion from the main text’s Appendix.

S17.1 Berry Phase and No-Go Results

The mean Berry phase over all 1200 triangular faces of the 600-cell, computed via $SO(3)$ parallel transport on S^3 projected through the Hopf fibration, is:

$$\langle \Omega_{\text{Berry}} \rangle = \frac{1}{\phi^4} = 3 - 2\phi = 0.14590 \quad (\text{S71})$$

Proposition S5 (Static geometry no-go). *The sector-dependent holonomy corrections (δ_d, δ_k) cannot arise from the static geometry of the 600-cell alone. Specifically:*

- (i) **Isotropic edge weights.** *The A_5 -irrep projector weights $w_R = d_R/12$ are identical for all 720 edges.*
- (ii) **Hopf-gauge independence.** *The Hopf fiber/base decomposition of edges is statistically independent of the gauge decomposition ($\chi^2 = 0$ exactly).*
- (iii) **Face-transitive holonomy.** *The $SO(3)$ holonomy is identical for all 1200 triangular faces. The mean Berry phase is $1/\phi^4$ universally.*

S17.2 McKay–CG Yukawa Structure

Proposition S6 (McKay–CG Yukawa structure). *The finite Dirac operator D_F on the McKay graph of $2I$ (affine $\hat{E}_8$, 9 nodes, 8 edges) satisfies:*

- (i) *All 16 Clebsch–Gordan intertwining maps have multiplicity 1 (unique Yukawa shape per edge).*
- (ii) *The Frobenius norm is universal: $\|Y_e\|_F = 1/\sqrt{2}$ for every edge e , independent of endpoint dimensions.*
- (iii) $\text{Tr}(D_F^2) = 8 = \text{rank}(E_8)$.
- (iv) *The Cayley-graph Laplacian eigenvalue per irrep ρ_i is $\lambda_i = 12(1 - \chi_i(10a)/d_i)$, giving the Galois-conjugate pair $\lambda(\mathbf{3}) = 12 - 4\phi$ and $\lambda(\mathbf{3}') = 8 + 4\phi$, with spectral gap $4\sqrt{a_1}$.*

The CG maps determine interaction selection rules, while mass eigenvalues require the $\mathbb{Z}[\phi]$ lattice.

S17.3 McKay–Cayley Spectral Sum Rules

Proposition S7 (McKay–Cayley spectral sum rules). *The adjacency matrix $\hat{A}$ of the McKay graph of $2I$ has $N_{\text{eig}} = 9$ nodes with dimensions $d_i = \{1, 2, 3, 4, 5, 6, 4, 2, 3\}$. Its spectrum is $\{0, \pm 1/\phi, \pm 1, \pm \phi, \pm 2\}$, and the Perron eigenvector equals the dimension vector exactly.*

The Cayley-graph Laplacian eigenvalue $\lambda_i = a_i + b_i\phi$ satisfies:

$$\sum_{i=1}^9 a_i = 12 = \dim(G), \quad (\text{S72})$$

$$\sum_{i=1}^9 b_i = 8 = \text{rank}(E_8), \quad (\text{S73})$$

$$\sum_{i=1}^9 n_i = 108 = N_{\text{eig}} \times \text{vertex degree}, \quad (\text{S74})$$

$$\sum_{i=1}^9 (a_i + b_i\phi) = 12 + 8\phi = 2/\alpha_s + \text{rank}(E_8). \quad (\text{S75})$$

The $2I$ character table is recovered exactly from the eigenvectors of $\hat{A}$.

S17.4 Galois Anti-Symmetric Cayley Eigenvalue

Proposition S8 (Galois anti-symmetric Cayley eigenvalue). *For each irrep ρ_i of $2I$, define the Galois anti-symmetric Cayley eigenvalue:*

$$G_i = \frac{\lambda_i(10a) - \lambda_i(5a)}{2} = \frac{6[\chi_i(5a) - \chi_i(10a)]}{d_i} \quad (\text{S76})$$

Then:

- (i) $G_i = 0$ for all WHITE/ A_5 irreps.

(ii) $G_i \neq 0$ for all BLACK/spin irreps.

(iii) The BLACK values are: $G(\rho_2) = -6\phi$, $G(\rho_4) = -3$, $G(\rho_6) = +2$, $G(\rho_8) = +6/\phi$.

(iv) $|G(\rho_4)| = 3 = N_{\text{gen}}$.

Proof. The McKay graph of $2I$ is bipartite. WHITE nodes carry A_5 representations with $\chi_i(5a) = \chi_i(10a)$, giving $G_i = 0$. BLACK nodes have $\chi_i(5a) = -\chi_i(10a)$, giving $G_i = -12\chi_i(10a)/d_i$. Explicitly: ρ_2 : $\chi(10a) = \phi$, $G = -6\phi$; ρ_4 : $\chi(10a) = 1$, $G = -3$; ρ_6 : $\chi(10a) = -1$, $G = +2$; ρ_8 : $\chi(10a) = -1/\phi$, $G = +6/\phi$. The symmetric part $S_i = 12$ universally for BLACK irreps. $\square$

S17.5 McKay Mass Correspondence

Theorem S5 (McKay mass correspondence). *On $\hat{E}_8$ rooted at ρ_1 , assigning fermion mass exponents as weighted graph distances, the main-chain weights are uniquely forced to $[N_{\text{gen}}, F(3), b_1, 0, a_1]$. Three solutions exist, distinguished only by branch assignments.*

Proof. The $\hat{E}_8$ tree has a unique path from ρ_1 to every node. A computer-assisted search over all $8!/2 = 20,160$ assignments confirms that $n(\rho_9) = 16$ (charm) is the only value admitting non-negative integer weights on all 8 edges, with main-chain sequence $\{3, 5, 11, 11\}$ forced. $\square$

Corollary S1 (Complementarity). *Three structures encode complementary aspects of the fermion spectrum: (i) D_F (selection rules), (ii) weighted McKay distance (hierarchy), (iii) $\mathbb{Z}[\phi]$ lattice (absolute masses).*

S17.6 Fermionic Lagrangian

The product geometry $\mathcal{M} \times F$ defines the complete spectral triple with:

$$\mathcal{H} = \mathcal{H}_M \otimes \mathcal{H}_F, \quad \dim = 2640 \times 30 = 79,200, \quad (\text{S77})$$

$$D = D_M \otimes \mathbf{1}_{30} + \gamma_{\text{form}} \otimes D_F, \quad (\text{S78})$$

$$\gamma = \gamma_{\text{form}} \otimes \gamma_F, \quad (\text{S79})$$

where $D_M = d + d^*$ is the simplicial Dirac operator, $\gamma_{\text{form}} = (-1)^p$ is the form-degree parity, and $\gamma_F = (-1)^{2j}$ is the bipartite chirality. The chirality axiom $\{\gamma, D\} = 0$, $\gamma^2 = 1$ is satisfied.

The chiral decomposition is balanced: $\dim(\mathcal{H}_+) = \dim(\mathcal{H}_-) = 39,600$, with $SU(2)$ coupling only to the BLACK sector (chiral) and $SU(3)$, $U(1)$ coupling to both (vector-like).

The finite Dirac operator decomposes as $D_F = \sum_e \phi^{w_e} D_e$, where D_e has off-diagonal blocks Y_{ij} (CG-derived Yukawa matrices) and w_e are the McKay distance weights.

The fermionic action is $S_F[\Psi] = \langle \Psi, D\Psi \rangle = S_{\text{kin}} + S_{\text{Yuk}}$.

Masses as Wilson lines. Physical masses are Wilson lines on the McKay tree:

$$\frac{m_f}{m_e} = \prod_{e \in \text{path}(\rho_1 \rightarrow \rho_f)} \phi^{w_e} = \phi^{n_f} \quad (\text{S80})$$

D_F encodes local coupling strengths, while Wilson lines integrate along the unique tree path. The mass operator $V = \sum_e w_e P_e$ does not commute with the adjacency operator: $[V, A] \neq 0$.

Comparison with standard NCG. In our framework: $\gamma_F = (-1)^{2j}$ is *derived* from bipartite structure; D_F has zero free parameters ($\|Y\|_F = 1/\sqrt{2}$); $N_{\text{gen}} = 3$ is derived. The manifold chirality $\gamma_{\text{form}} = (-1)^p$ is mathematically valid but the continuum limit identification with γ_5 remains open.

Open. (a) Chiral $SU(2)$ coupling is derived from the $\mathbb{Z}/2$ ring-homomorphism. (b) The μ - s degeneracy forces one T_3 mismatch (8/9 correct). (c) Continuum limit convergence is not proven.

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